

Chapter 6

Dynamics of a Rigid Body

6.1 Rotation about a fixed axis and rotational inertia

Rigid solid $\sim$ rotation

general rotation $\sim$ rotation about a fixed axis

$$\boldsymbol{\omega} = (\omega_1, \omega_2, \omega_3)$$

$$\boldsymbol{p} = m\boldsymbol{v}$$

$$\boldsymbol{L} = ? \boldsymbol{\omega}$$

$$\mathbf{L} = \mathbf{r} \times m\mathbf{v}$$

$$= m(\rho \mathbf{e}_\rho + z\mathbf{k}) \times [\omega \mathbf{k} \times (\rho \mathbf{e}_\rho + z\mathbf{k})]$$

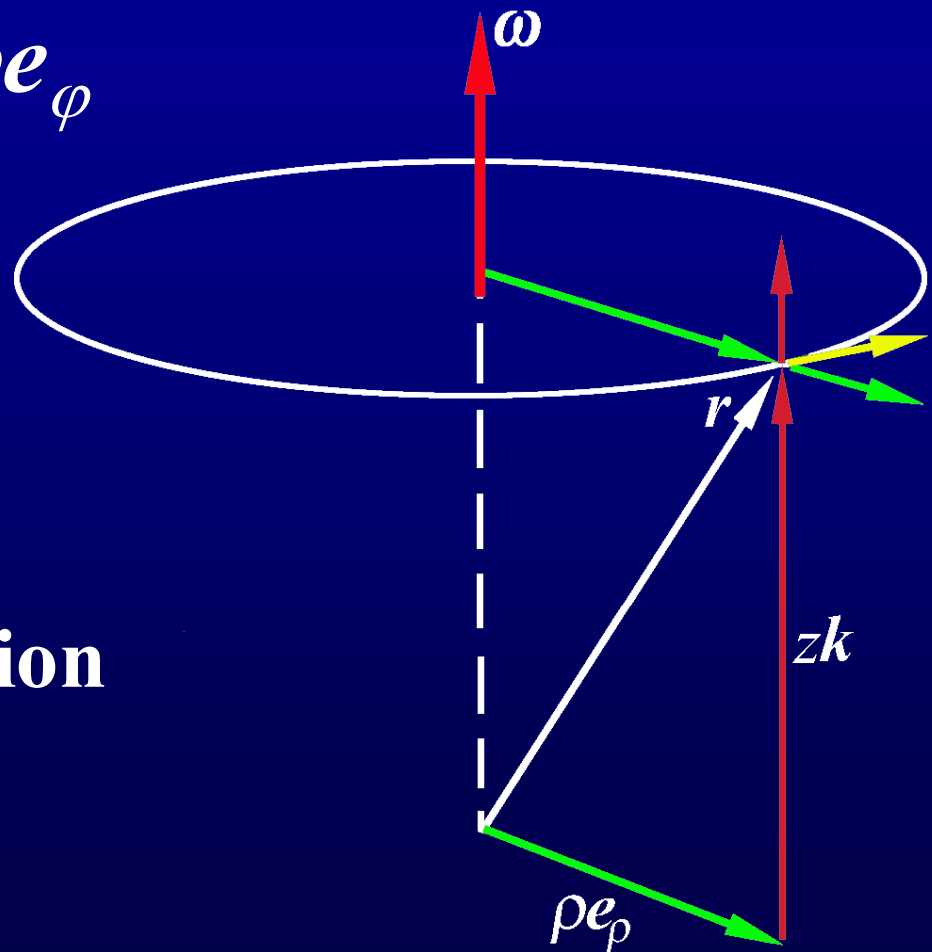
$$= m(\rho \mathbf{e}_\rho + z\mathbf{k}) \times \omega \rho \mathbf{e}_\phi$$

$$= m\rho^2 \omega - \omega m z \rho$$

For a **single particle**
we may set $z = 0$

We have a simple relation

$$\mathbf{L} = m\rho^2 \omega$$



For a system of particles,

$$L = \sum_i m_i \rho_i^2 \omega - \omega \sum_i m_i z_i \rho_i$$

or for continuous distribution of mass

$$L = \left(\int \rho^2 dm \right) \omega - \omega \left(\int z \rho dm \right)$$

It is still possible to set $z = 0$ if all m_i s are in motion plane.

In general, it is impossible to set $z = 0$ simply.

If $\int z\rho dm = 0$ (valid for **symmetric axis**
or **principal axis of inertia**)

we still have a linear relation similar to
relation for a single particle

$$L = \left(\int \rho^2 dm \right) \omega$$

where

$$I \equiv \int \rho^2 dm$$

is called **rotational inertia** or **moment of inertia**

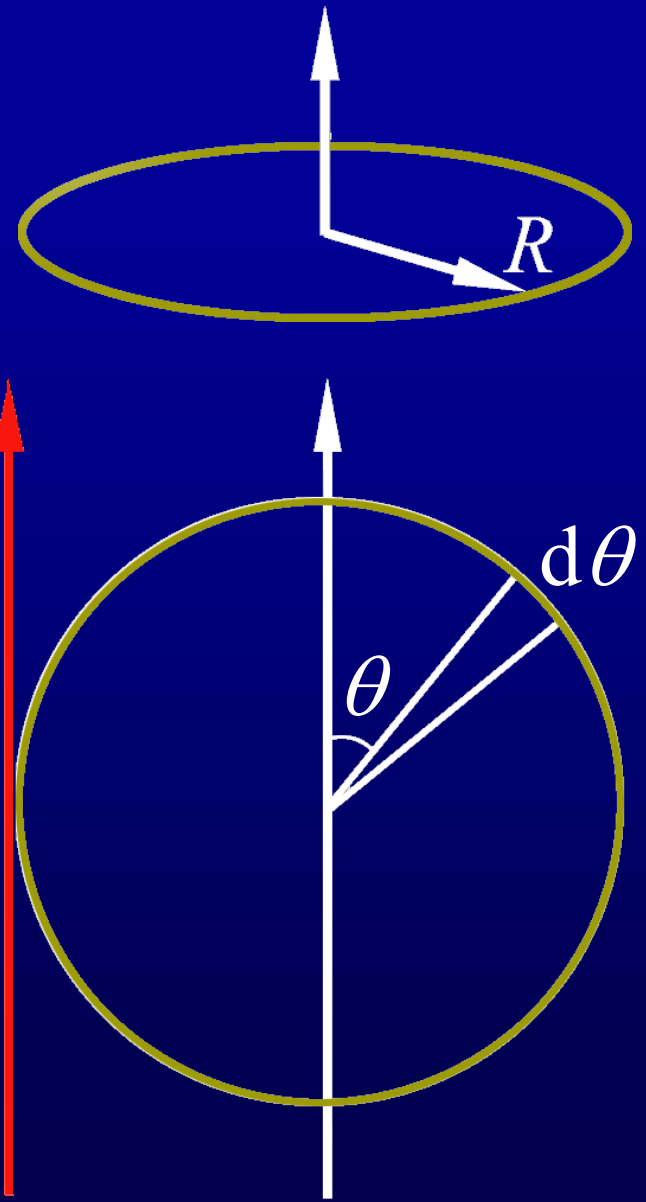
Example 6.1

$$I = \int R^2 dm = mR^2$$

$$I = 2 \int_0^{\pi} (R \sin \theta)^2 \frac{m}{2\pi R} R d\theta$$

$$= mR^2 \frac{1}{\pi} \int_0^{\pi} \sin^2 \theta d\theta$$

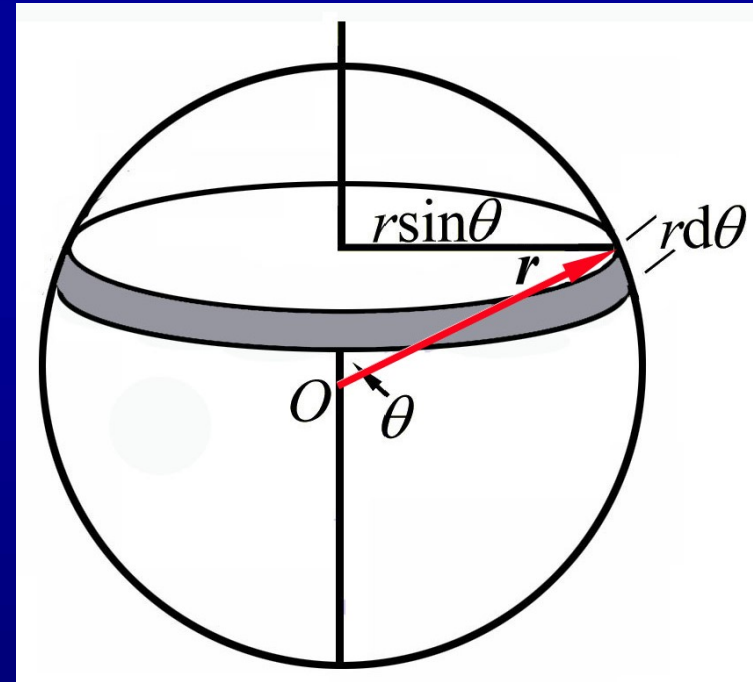
$$= \frac{1}{2} mR^2$$



$$I = \int (r \sin \theta)^2 dm$$

$$= \iint (r \sin \theta)^2 \frac{m}{\frac{4}{3}\pi R^3} [2\pi (r \sin \theta) r d\theta dr]$$

$$= \frac{3m}{2R^3} \int_0^R r^4 dr \int_0^\pi \sin^3 \theta d\theta = \frac{2}{5} mR^2$$



The radius of gyration R_0 is defined through

$$I \equiv mR_0^2$$

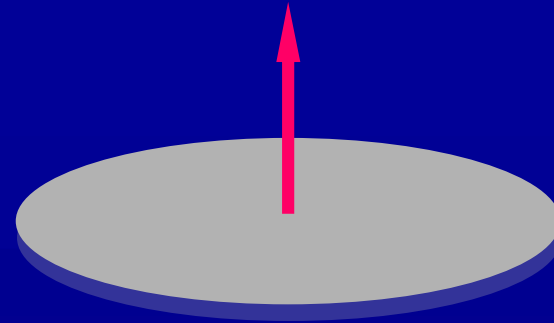
In Example 6.1

$$I' = mR^2, \quad I'' = \frac{1}{2}mR^2$$

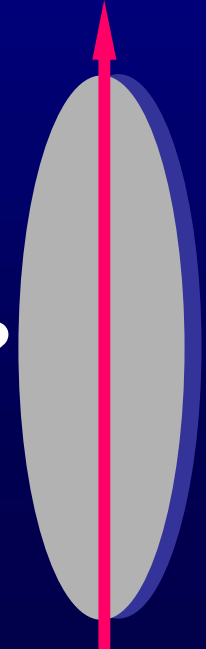
$$R'_0 = R, \quad R''_0 = \frac{1}{\sqrt{2}}R$$

For a disc

$$\begin{aligned} I &= \int_0^R r^2 dm \\ &= \int_0^R r^2 \frac{m}{\pi R^2} 2\pi r dr \\ &= \frac{m}{R^2} \int_0^R r^2 dr^2 \\ &= \frac{1}{2} m R^2 \end{aligned}$$



What about a cylinder?
Around a diameter?

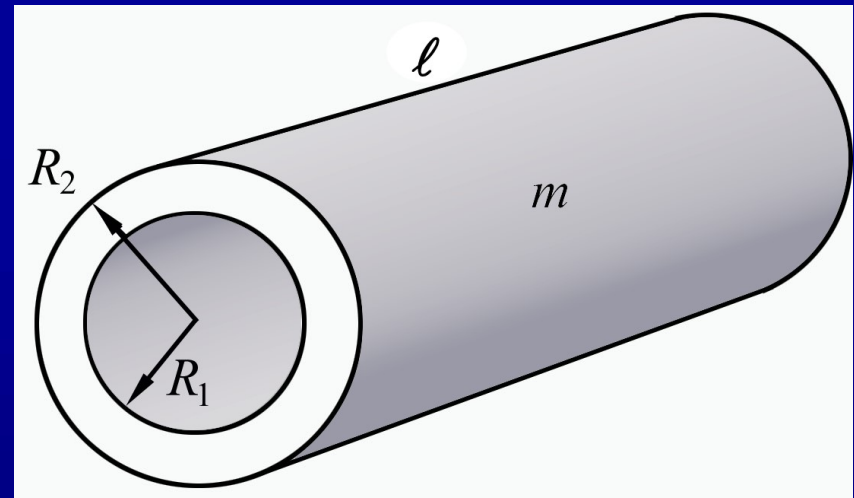


Example 6.2

Cylindrical shell

$$I \sim \frac{1}{2} m_2 R_2^2 - \frac{1}{2} m_1 R_1^2 ?$$

Solution:

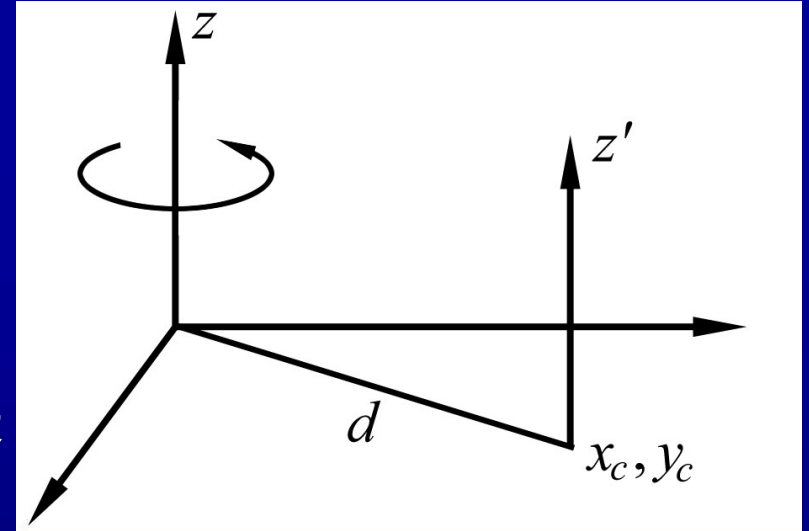


$$\begin{aligned} dm &= \rho (2\pi r) dr \ell \\ &= \frac{m}{\pi (R_2^2 - R_1^2) \ell} 2\pi r dr \ell = \frac{2m}{R_2^2 - R_1^2} r dr \end{aligned}$$

$$I = \int r^2 dm = \frac{1}{2} m (R_2^2 + R_1^2)$$

parallel axis theorem

$$\begin{aligned} I &= \int (x^2 + y^2) dm \\ &= \int [(x' + x_C)^2 + (y' + y_C)^2] dm \\ &= \int (x'^2 + y'^2) dm + (x_C^2 + y_C^2) \int dm \\ &= I_C + md^2 \end{aligned}$$



where

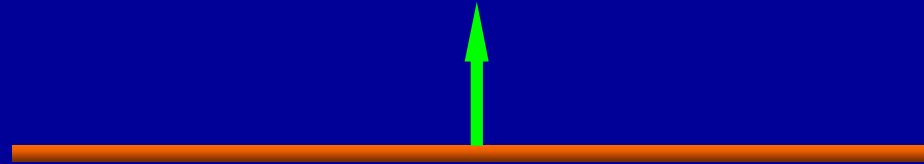
$$2x_C \int x' dm = ?$$

$$\sum_i m_i \mathbf{r}_i = m_C \mathbf{r}_C$$

$$2x_C m_C x'_C \longrightarrow 0$$

Example 6.3

$$I = \int_{-\frac{\ell}{2}}^{\frac{\ell}{2}} x^2 \boxed{\frac{m}{\ell} dx} = \frac{1}{12} m \ell^2$$

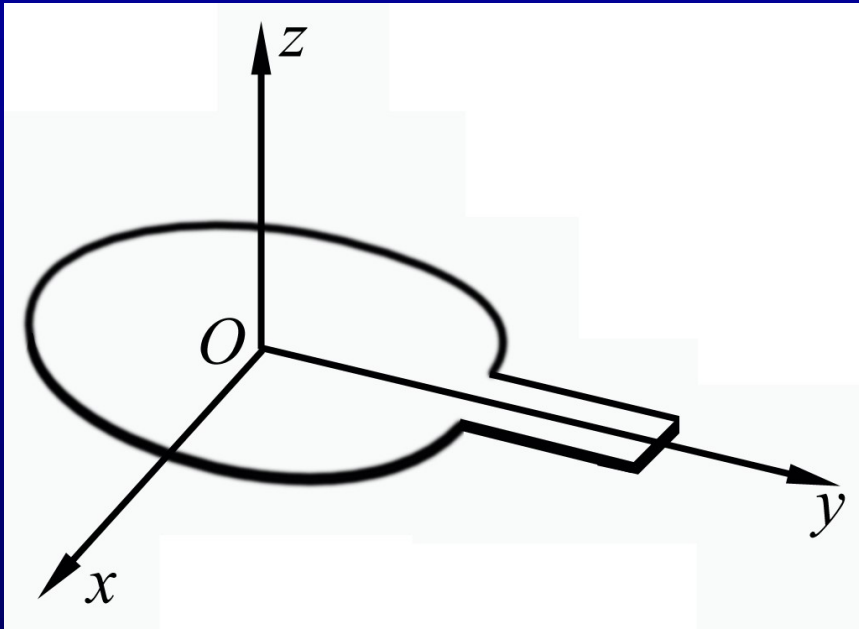


$$I = \frac{1}{12} m \ell^2 + m(\ell/2)^2 = \frac{1}{3} m \ell^2$$

I_C

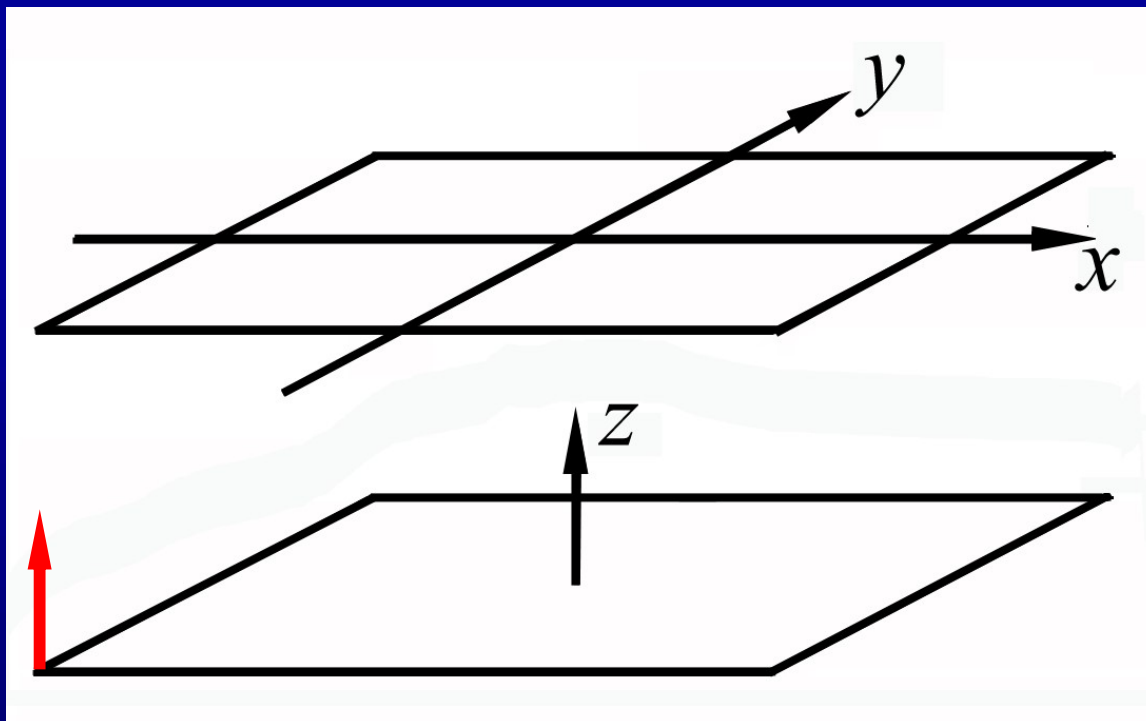
" md^2 "

perpendicular axis theorem



$$\begin{aligned} I_z &= \int dm(x^2 + y^2) \\ &= \int dm x^2 + \int dm y^2 \\ &= I_y + I_x \end{aligned}$$

valid for thin plate.



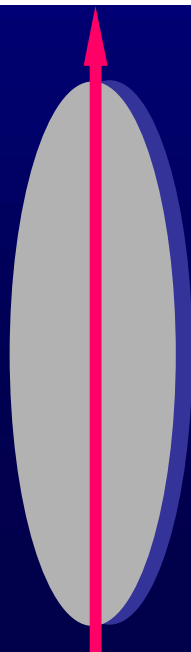
$$I_x = \frac{1}{12} m \ell^2$$

$$I_y = \frac{1}{12} m b^2$$

$$I_z = I_x + I_y$$

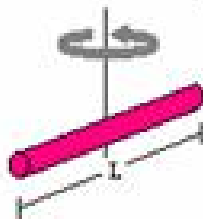
$$= \frac{1}{12} m (\ell^2 + b^2)$$

Remember



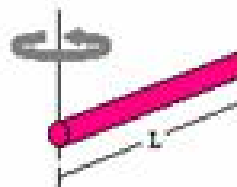
$$\frac{1}{2} m R^2 = I_d + I_d$$

Long thin rod with
rotation axis through center



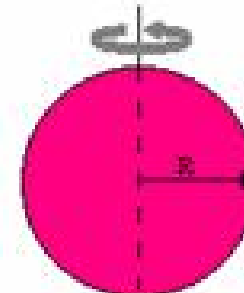
$$I = \frac{1}{12} ML^2$$

Long thin rod with
rotation axis through end



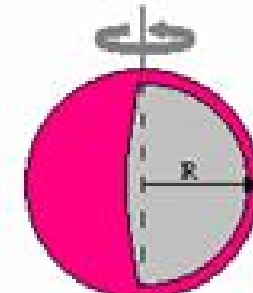
$$I = \frac{1}{3} ML^2$$

Solid sphere



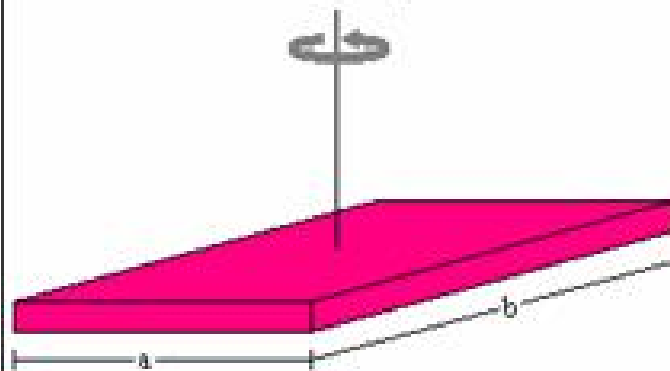
$$I = \frac{2}{5} MR^2$$

Thin spherical shell



$$I = \frac{2}{3} MR^2$$

Rectangular plate



$$I = \frac{1}{12} M(a^2 + b^2)$$

Hoop or cylindrical shell



$$I = MR^2$$

Hollow cylinder



$$I = \frac{1}{2} M(R_1^2 + R_2^2)$$

Solid cylinder



$$I = \frac{1}{2} MR^2$$

6.2 Dynamics of Rotation

$3N$ coordinates ($\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_N$)

constraints $|\mathbf{r}_i - \mathbf{r}_j| = C_{ij}$

$$C_N^2 = N(N-1)/2 \quad \textbf{pairs}$$

Independent coordinates $3N - N(N-1)/2$

1 reference point fixed

2 reference point fixed

3 reference point fixed (non-collinear)

3 points → 9 coordinates; 3 constraints
 $3 \times 3 - 3 = 6$ Independent variables
six degrees of freedom

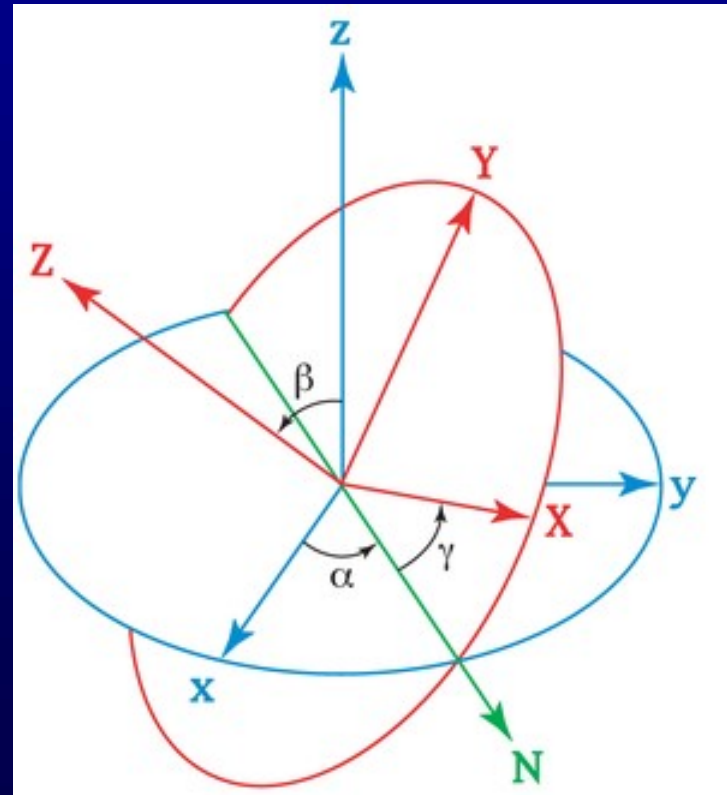
Position : r_C

Orientation:

Euler angles : $\alpha \quad \beta \quad \gamma$

$$m_C \frac{d^2 r_C}{dt^2} = F_{\text{ext}}$$

$$\frac{dL}{dt} = M$$



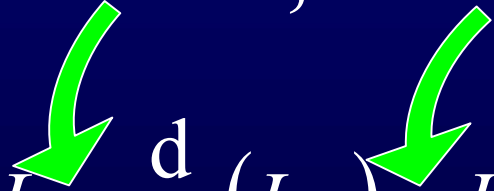
$$\frac{d\mathbf{L}}{dt} = \mathbf{M}$$

If $\mathbf{M} \parallel \mathbf{L}$, $d\mathbf{L} \sim \mathbf{L}$

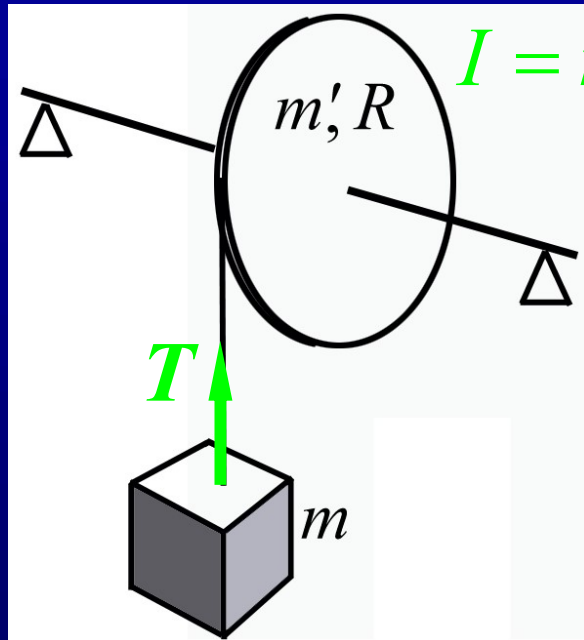
*If $\mathbf{M} \perp \mathbf{L}$, $d\mathbf{L} \perp \mathbf{L}$ (*precession*)*

Arbitrary $\mathbf{M} = \mathbf{M}_{\parallel} + \mathbf{M}_{\perp}$

For symmetric, fixed axis

$$\mathbf{M} = \frac{d}{dt} \mathbf{L} = \frac{d}{dt} (I\omega) = I \frac{d}{dt} \omega \equiv I\alpha$$


Example 6.4



$$I = m'R^2/2$$

(a) 9.8N:

$$\sum F = 0 \text{ (axis \& disc)}$$

$$RF = \left(\frac{1}{2} m'R^2 \right) \alpha$$

$$\alpha = \frac{2F}{m'R}$$

(b) 1.0kg:

$$mg - T = m(R\alpha) \quad (m)$$

9.8N or
1.0kg

For axis & ...

$$RT = \left(\frac{1}{2} m'R^2 \right) \alpha$$

$$\alpha = \frac{2mg}{(m' + 2m)R}$$

rolling down an incline

$$x: mg \sin \theta - F_f = ma$$

$$y: F_N - mg \cos \theta = 0$$

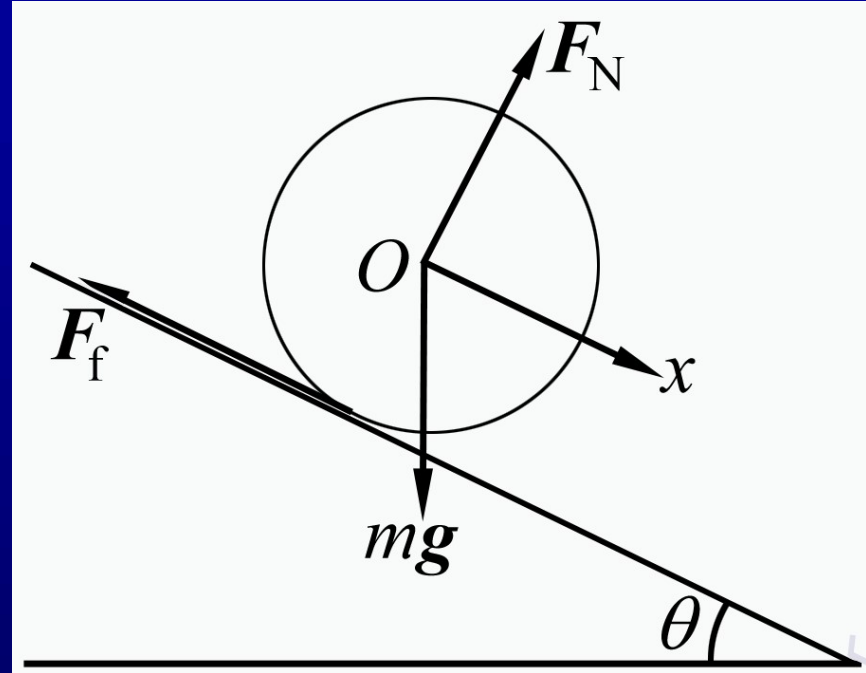
$$R F_f = \left(\frac{1}{2} m R^2 \right) \alpha$$

$$R \alpha = a$$

$$a = \frac{2}{3} g \sin \theta = C; \quad v^2 = 2as = \frac{4}{3} g s \sin \theta$$

$$F_f = \frac{1}{3} mg \sin \theta, \quad \mu \geq \frac{1}{3} \tan \theta \quad \text{rolling without slipping}$$

Planar motion of a rigid body



Resultant torque of gravity forces on a rigid body

The gravity force on dm is

$$d\mathbf{F} = dm(-g\mathbf{k})$$

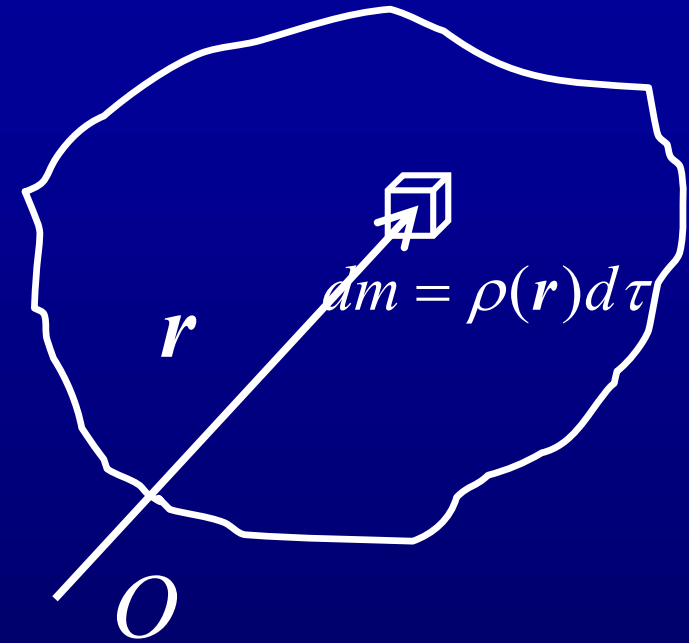
$$\left(\begin{array}{l} \mathbf{g} = -g\mathbf{k} = \text{Const.} \\ \text{--- uniform gravitational field} \end{array} \right)$$

The resultant gravity force is

$$\mathbf{F} = \int d\mathbf{F} = \int dm(-g\mathbf{k}) = -mg\mathbf{k}.$$

And the resultant torque is

$$\begin{aligned} \mathbf{M} &= \int \mathbf{r} \times d\mathbf{F} = \int \mathbf{r} \times dm(-g\mathbf{k}) = \left(\int \mathbf{r} dm \right) \times (-g\mathbf{k}) \\ &= \frac{\int \mathbf{r} dm}{m} \times m(-g\mathbf{k}) = \mathbf{r}_C \times (-mg\mathbf{k}). \end{aligned}$$



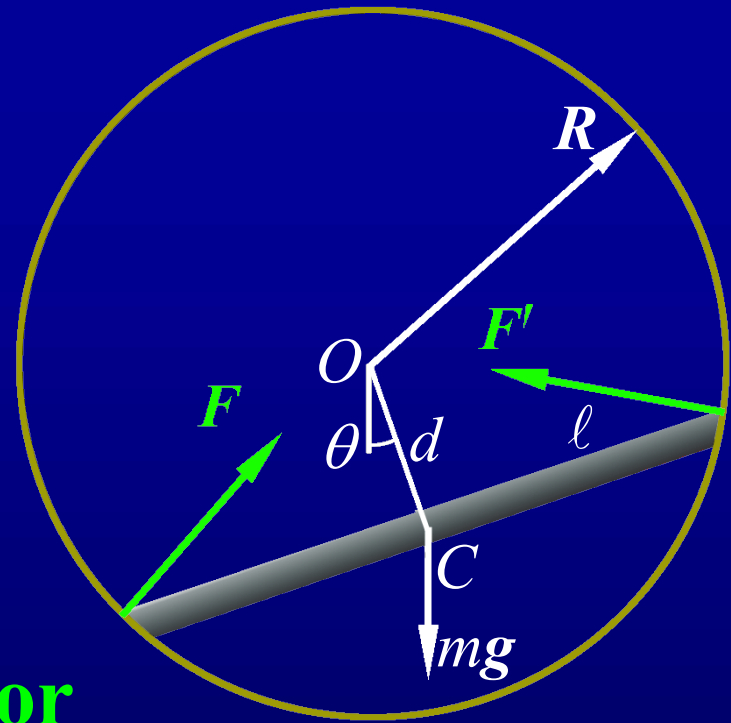
Example 6.5

How many coordinates we need to determine the position of the rod?

Step 1, decide the point torque **with respect to: C or O ?**

$$-mg(d \sin \theta) = I_O \ddot{\theta}$$

Step 2, find the rotational inertia about O .



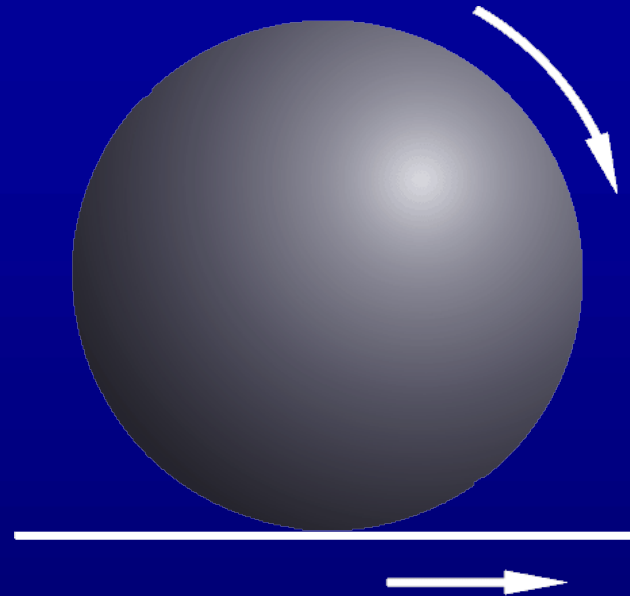
Example 6.6

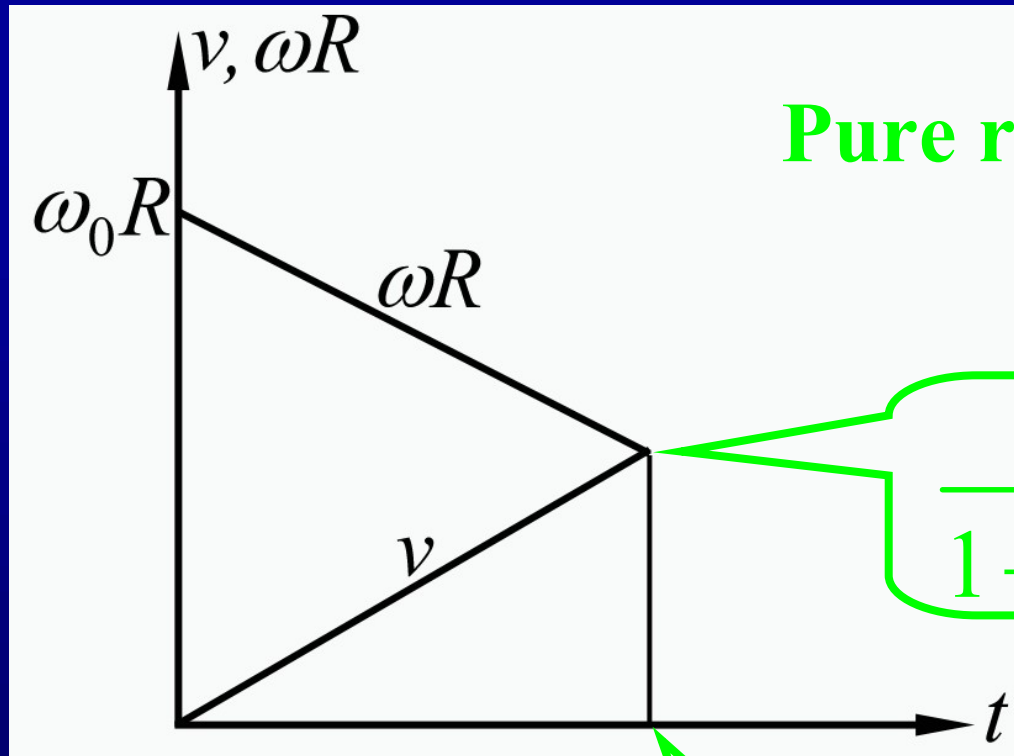
$$\mu_k mg = m \frac{dv}{dt} \quad (1)$$

$$- R \mu_k mg = I \frac{d\omega}{dt} \quad (2)$$

$$(1): v = \mu_k gt$$

$$\begin{aligned} (2): \omega &= \omega_0 - \frac{mR}{I} \mu_k gt \\ &= \omega_0 - \frac{mR}{I} v \end{aligned}$$





Pure rolling begins when

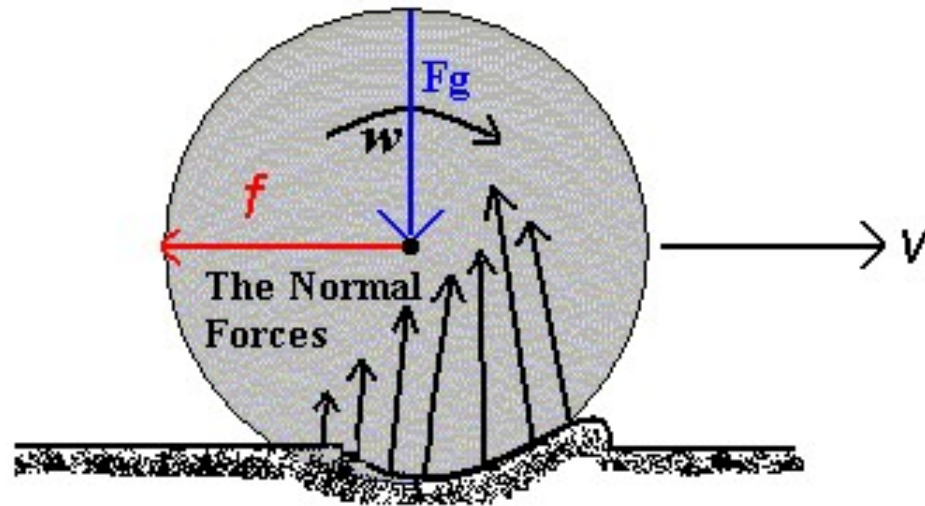
$$\omega R = v$$

$$\frac{\omega_0 R}{1 + mR^2/I}$$

$$v/\mu_k g$$

$$I = mR^2, \frac{1}{2}mR^2, \frac{1}{2}mR^2, \frac{2}{5}mR^2, \dots$$

Rolling Resistance (Friction)



The actual forces acting on the wheel and the surface. As one can see in this exaggerated view, both the wheel and the surface undergo deformation to an extent determined by the elastic properties of the two surfaces.

$$f_r = \frac{\mu_r}{R} N$$

Rotational kinetic energy

$$T = \frac{1}{2} m_C v_C^2 + \frac{1}{2} \sum_i m_i v_i^2$$

$$\frac{1}{2} \sum_i \Delta m_i v_i^2 = \frac{1}{2} \sum_i \Delta m_i \mathbf{v}_i \cdot (\boldsymbol{\omega} \times \mathbf{r}_i)$$

$$= \frac{1}{2} \boldsymbol{\omega} \cdot \left(\sum_i \mathbf{r}_i \times \Delta m_i \mathbf{v}_i \right) = \frac{1}{2} \boldsymbol{\omega} \cdot \mathbf{L}$$

For **principal axis** $\mathbf{L} = I \boldsymbol{\omega}$

$$\frac{1}{2} m_C v_C^2 + \frac{1}{2} I \omega^2 + U = \text{constant}$$

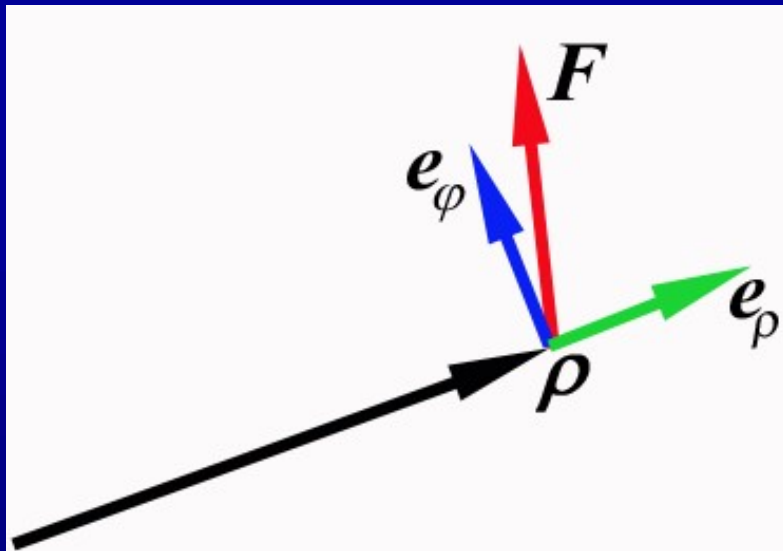
rolling down an incline

$$mgh = \frac{1}{2}mv_C^2 + \frac{1}{2}I\omega^2 + mg \cdot 0$$

$$= \frac{1}{2}mv_C^2 + \frac{1}{2}I \frac{v_C^2}{R^2}$$

$$v_C^2 = \frac{2gh}{1 + I/mR^2}$$

$$gh, \quad \frac{4}{3}gh, \quad \frac{10}{7}gh$$



work

$$\begin{aligned}
 dW &= \mathbf{F} \cdot (\rho d\phi \mathbf{e}_\phi) \\
 &= \rho F \cos(\hat{\mathbf{F}} \mathbf{e}_\phi) d\phi \\
 &= \rho F \sin(\mathbf{e}_\rho \hat{\mathbf{F}}) d\phi \\
 &= M d\phi
 \end{aligned}$$

power

$$P = \frac{dW}{dt} = M \frac{d\phi}{dt} = M\omega = M \cdot \omega$$

$$\sim P = \mathbf{F} \cdot \mathbf{v}$$

?

Assignment: 6.1, 6.2, 6.3

For the principal axes of inertia, please see the pages 190-196 in

Walter Greiner, *Classical Mechanics: Systems of Particles and Hamiltonian Dynamics*, Springer(2003).

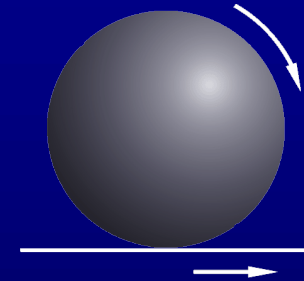
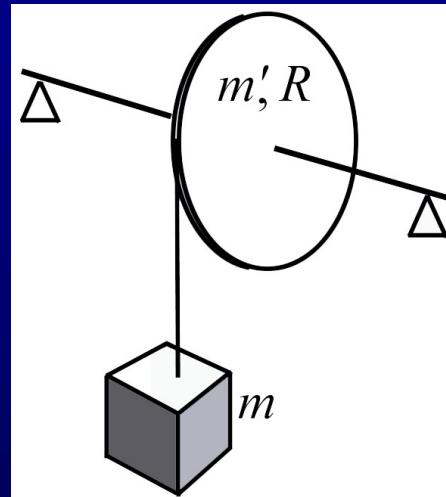
6.3 Precession of angular momentum

进动

$$\frac{d\mathbf{L}}{dt} = \mathbf{M} \quad d\mathbf{L} \sim \mathbf{M}$$

If $\mathbf{M} \sim \pm \mathbf{L}$, e.g.

$$d\mathbf{L} \sim \mathbf{M} // \pm \mathbf{L}.$$

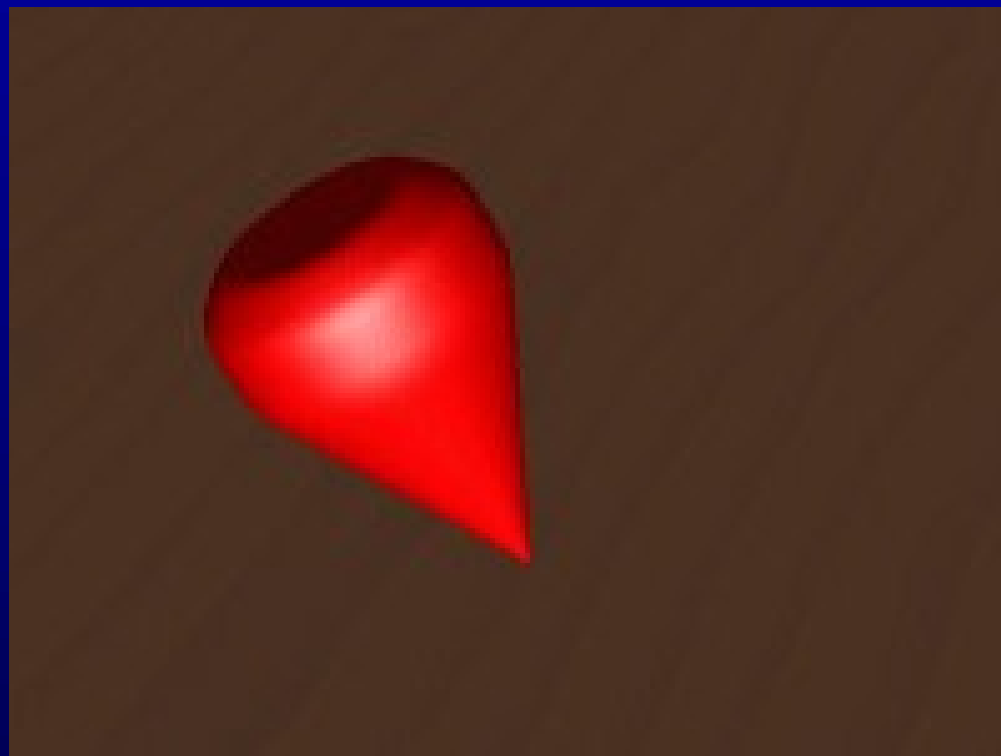


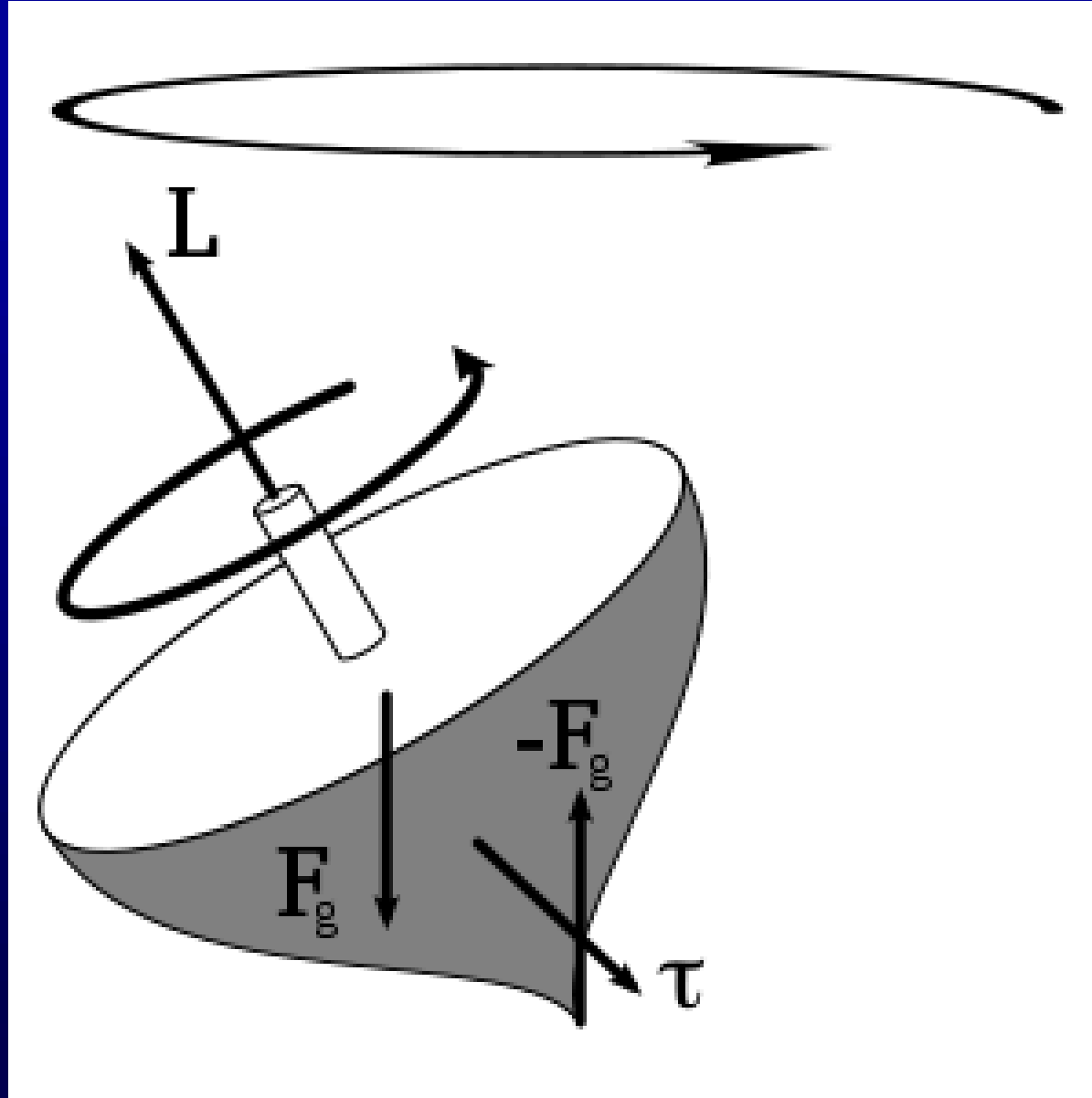
What will happen (to $d\mathbf{L}$) if $\mathbf{M} \perp \mathbf{L}$ or $\mathbf{M}$ has a perpendicular component?



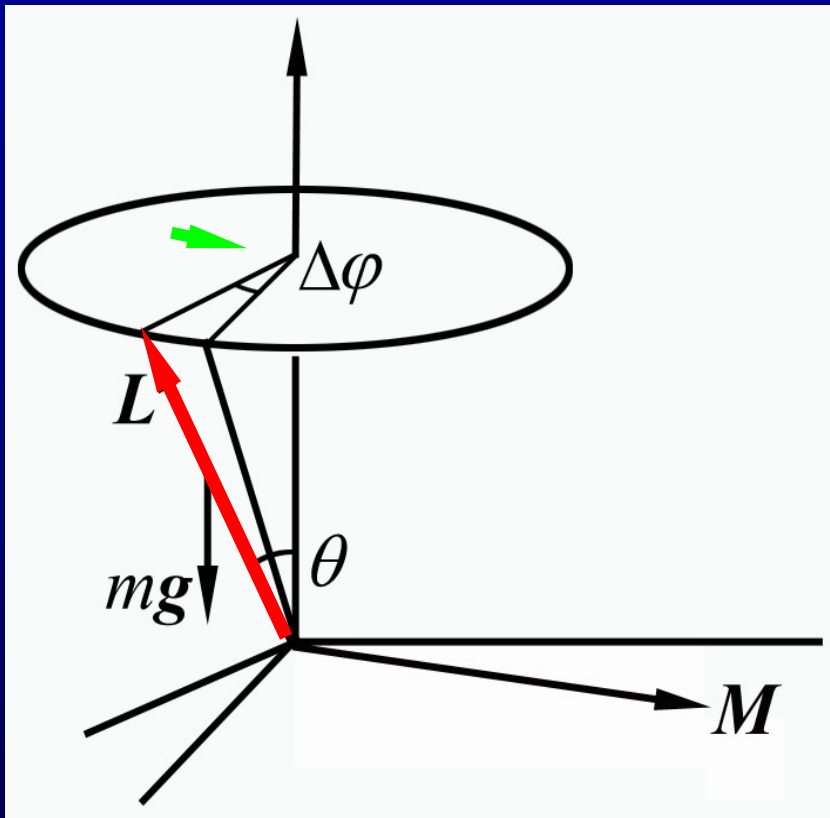
Peg-top







Symmetric top



$$|M| = |r \times mg| = rmg \sin \theta$$

$$\Delta L = L \sin \theta \Delta \phi$$

$$= I \omega \sin \theta \Delta \phi$$

$$\Omega = \lim_{\Delta t \rightarrow 0} \frac{\Delta \phi}{\Delta t}$$

$$= \lim_{\Delta t \rightarrow 0} \frac{\Delta L}{\Delta t} \frac{1}{I \omega \sin \theta}$$

$$= \frac{rmg \sin \theta}{I \omega \sin \theta}$$

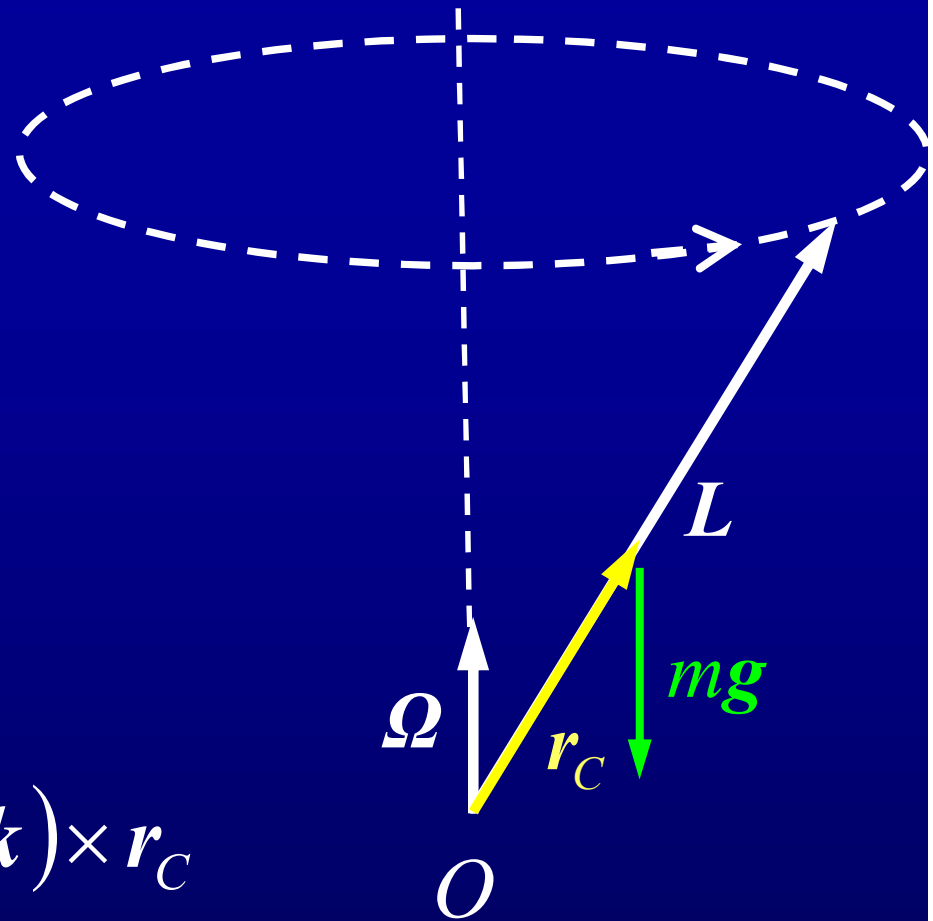
$$\frac{d\mathbf{L}}{dt} = \boldsymbol{\Omega} \times \mathbf{L}$$

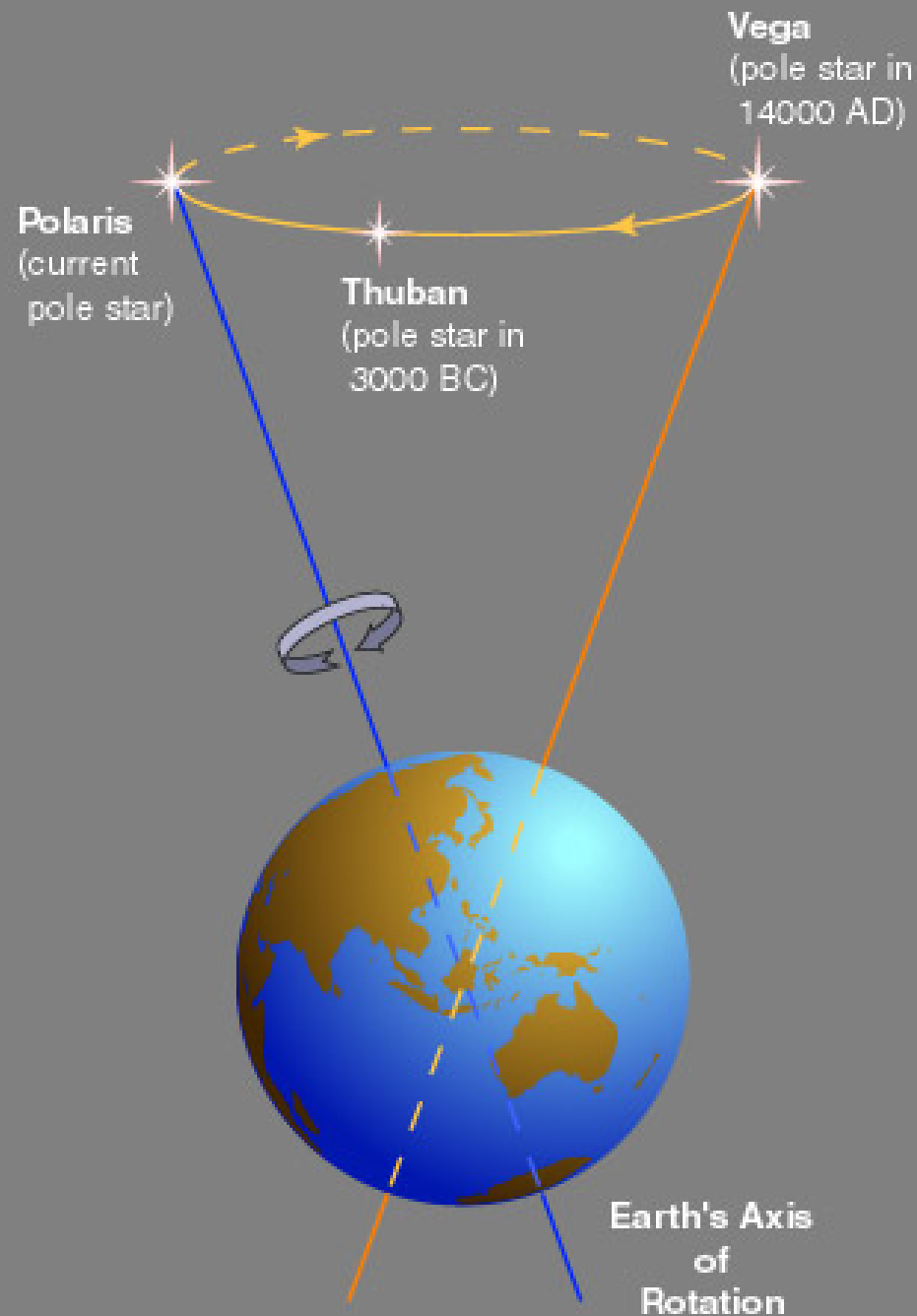
$$\mathbf{M} = \mathbf{r}_C \times m\mathbf{g}$$

$$\frac{d\mathbf{L}}{dt} = \mathbf{M} \Rightarrow$$

$$\boldsymbol{\Omega} \times \mathbf{L} = \mathbf{r}_C \times m\mathbf{g} = (mg\mathbf{k}) \times \mathbf{r}_C$$

$$\boldsymbol{\Omega} = \frac{mgr_C}{L} \mathbf{k} = \frac{mgr_C}{I\omega} \mathbf{k}$$





Polaris 北极星

Thuban

右枢, 紫微右垣(天龙座 α)

Vega 织女星
(Altair 牵牛星)

The equinoxes will
change in time too.

ecliptic

黄道

celestial

equator

天球赤道

equinox

分点

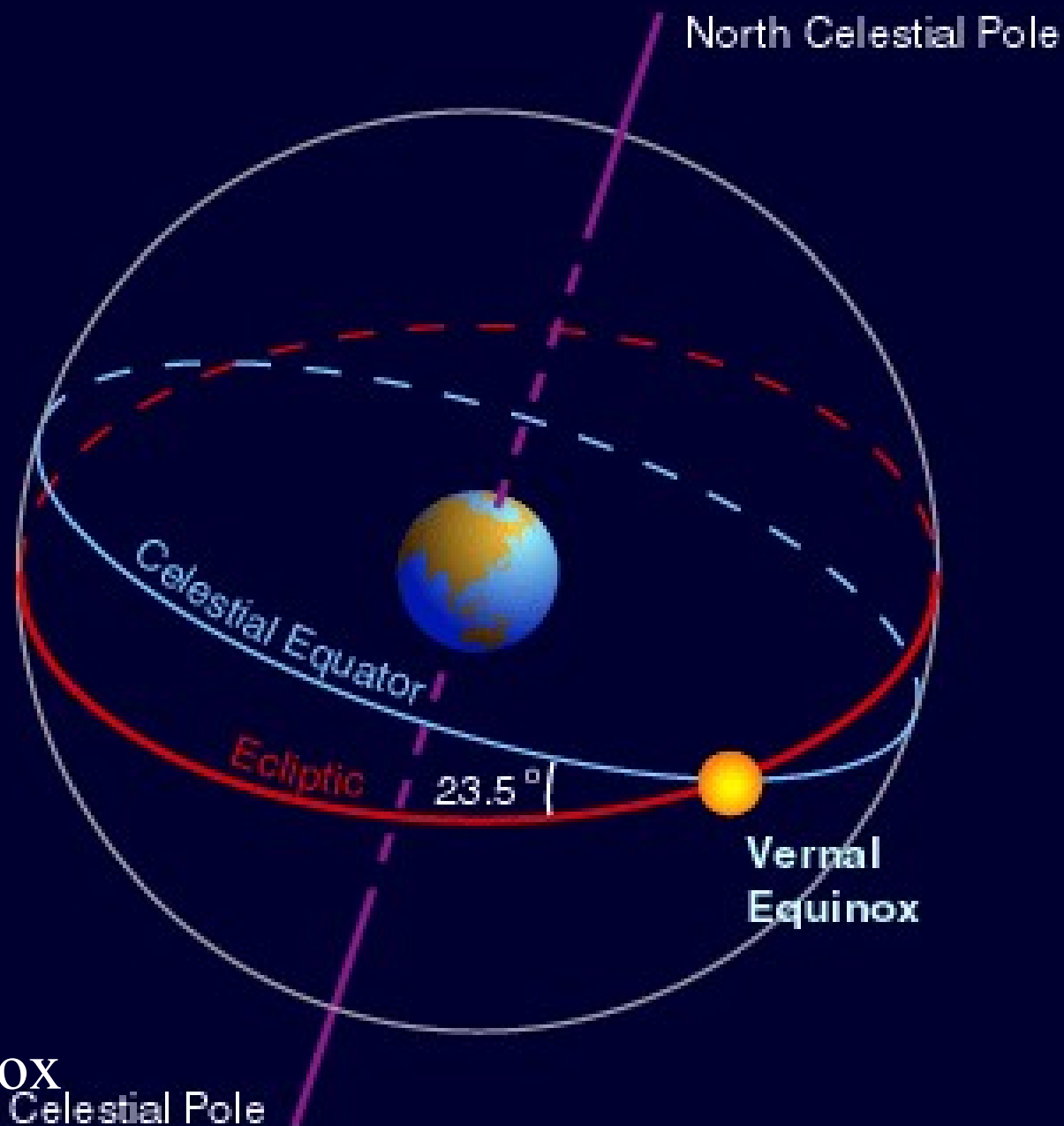
solstice

至点

vernal equinox

autumnal equinox

South Celestial Pole





Long exposure of stars appearing to circle the North Celestial pole. The daily rotation of Earth creates the appearance of rings around the pole.

precession of equinoxes

分点的进动或岁差

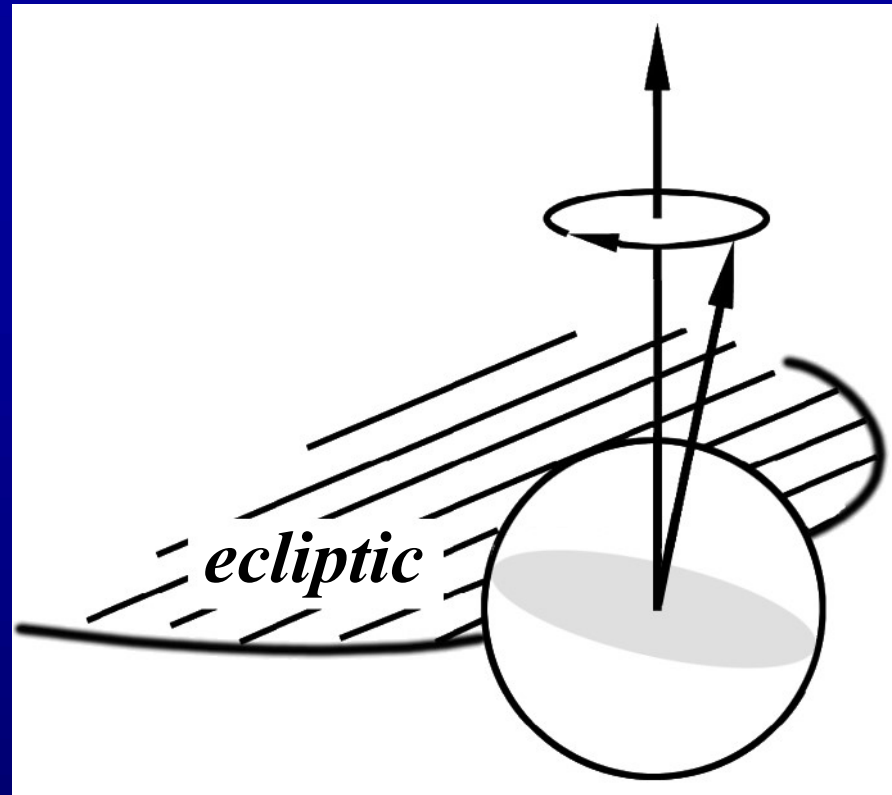
$$T = 27725\text{a} \quad \text{or}$$

$$\Omega = 7.19 \times 10^{-12} \text{ rad/s}$$

$$\omega = 7.292 \times 10^{-5} \text{ rad} \cdot \text{s}^{-1}$$

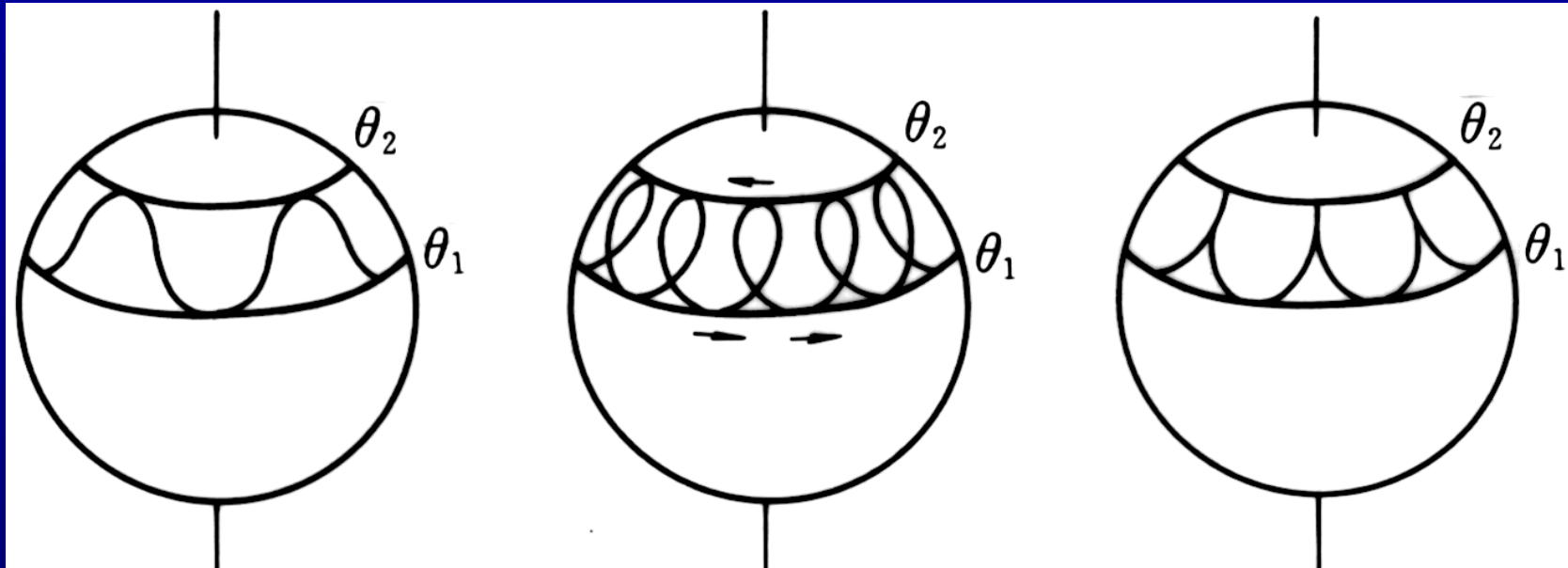
$$\begin{aligned} I &\approx \frac{2}{5} m R^2 = \frac{2}{5} (5.96 \times 10^{24} \text{ kg}) (6.38 \times 10^6 \text{ m})^2 \\ &= 9.74 \times 10^{37} \text{ m}^2 \cdot \text{kg} \end{aligned}$$

$$M = \Omega L \sin \theta = 2.03 \times 10^{22} \text{ N} \cdot \text{m}$$

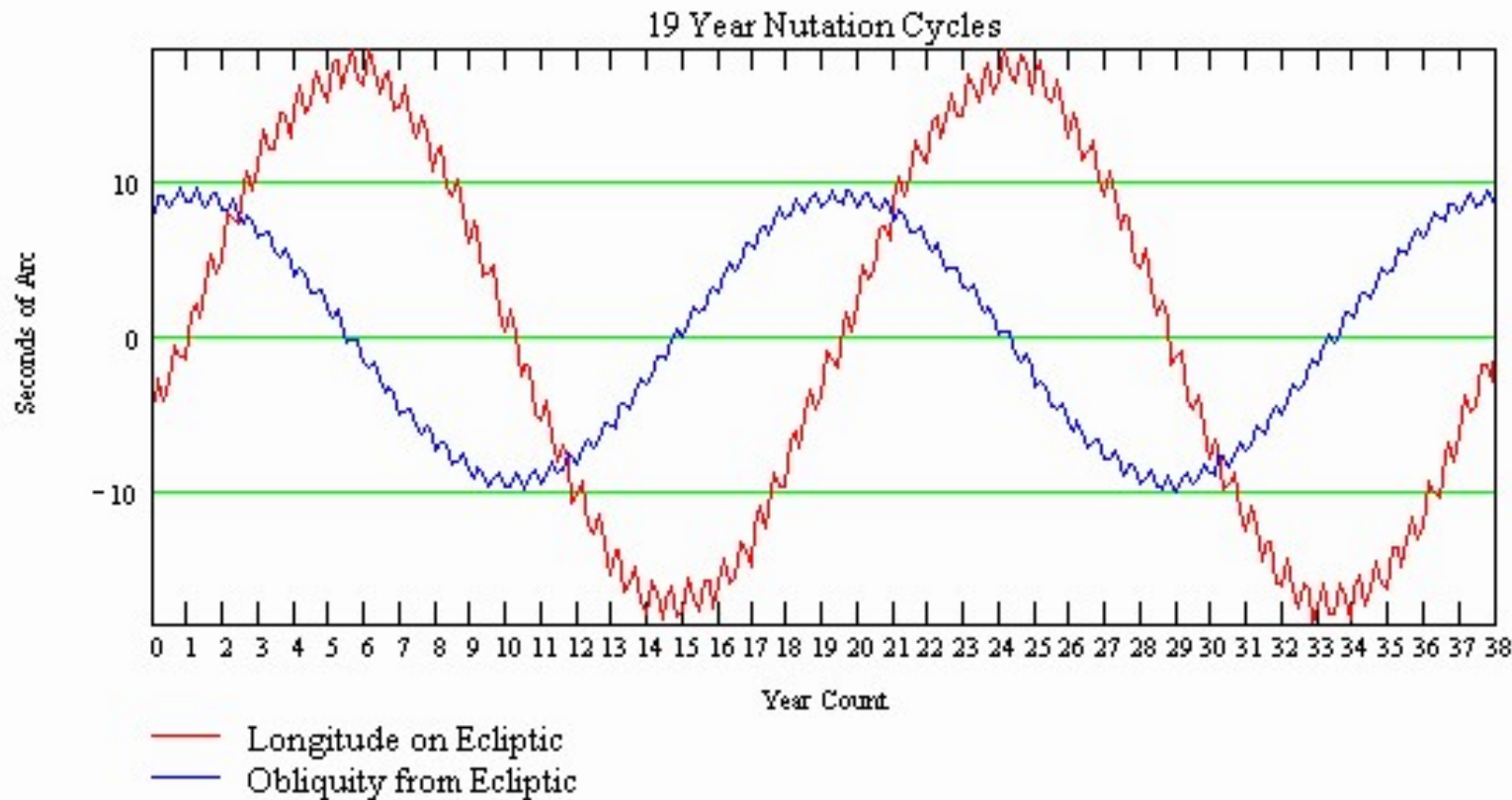


nutration

章动



Values of nutation are usually divided into components parallel and perpendicular to the ecliptic. The component which works along the ecliptic is known as the *nutration in longitude*(黄经章动). The component perpendicular to the ecliptic is known as the *nutration in obliquity*(交角章动).



Nutation is a small cyclical motion superimposed upon the steady 26,000-year precession of the Earth's axis of rotation. It is mainly caused by the gravitational effect of the 18.6-year rotation period of the Moon's orbit.

http://www.pietro.org/Astro_Util_StaticDemo/MethodsNutationVisualized.htm

6.4 Equilibrium of rigid bodies and stability

$$\sum F_{\text{ext}} = 0 \quad \text{translational equilibrium}$$

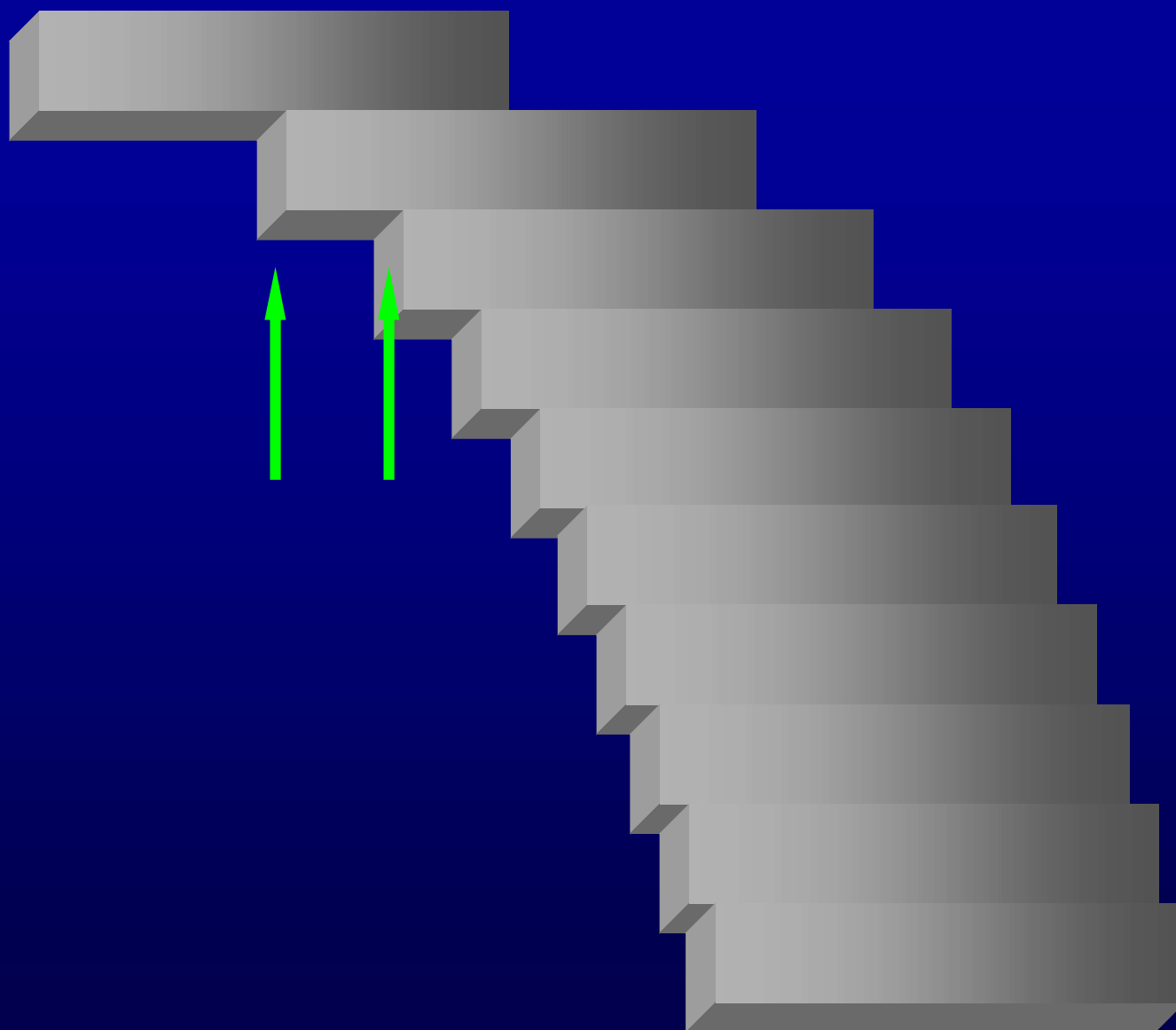
$$\sum M = 0 \quad \text{rotational equilibrium}$$

$$\longrightarrow a_C = 0, \quad \alpha = 0$$

mechanical equilibrium

$$v_C = 0, \quad \omega = 0$$

static equilibrium



Center of mass & center of gravity

$$\sum_i \mathbf{r}_i m_i = \mathbf{r}_C \sum_i m_i$$

We try to define COG through *

$$\sum_i \mathbf{r}_i \times (m_i \mathbf{g}_i) = \mathbf{r}_{\text{cog}} \times \sum_i m_i \mathbf{g}_i$$

But it is not completely fixed (the end point on a line).

Take the center of force as the origin,

$$\sum_i m_i \mathbf{g}_i = -G \frac{m_{\oplus} \sum_i m_i}{r_{\text{cog}}^3} \mathbf{r}_{\text{cog}}$$

Offers a radius from the center of force.

$$\frac{\left| \sum_i m_i \mathbf{g}_i \right|}{g \sum_i m_i} = \frac{R_{\oplus}^2}{r_{\text{cog}}^2}$$

In a uniform gravitational field or the dimension is much smaller than the radius of the earth, the COG and the COM coincide.

Unlike COM, COG depends on relative position and orientation. The relative character of the COG makes it of little use.

$$U' = 0$$

equilibrium

$$U'' \begin{cases} > 0 \\ = 0 \\ < 0 \end{cases}$$

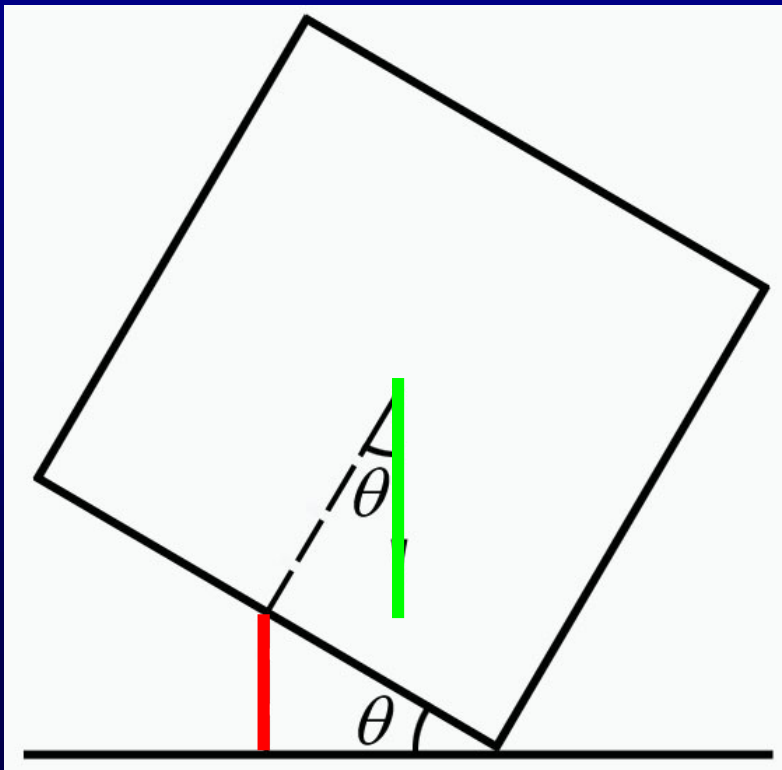
stable

neutral

unstable

Example 6.9

$$U(\theta) = \begin{cases} mg \left(\frac{a}{2} \cos \theta + \frac{a}{2} \sin \theta \right), & 0 \leq \theta \leq \frac{\pi}{2} \\ mg \left(\frac{a}{2} \cos \theta - \frac{a}{2} \sin \theta \right), & 0 \geq \theta \geq -\frac{\pi}{2} \end{cases}$$

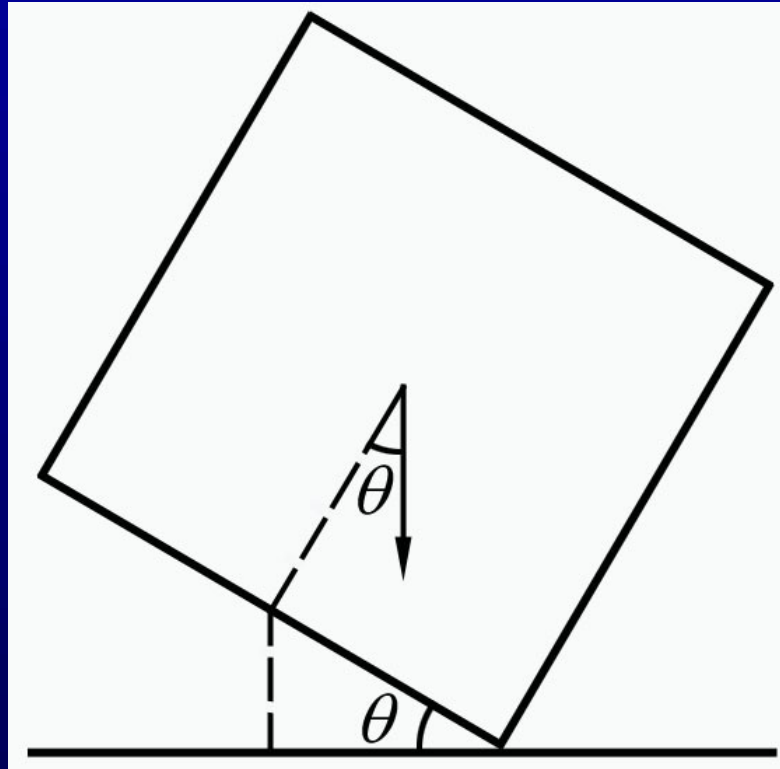


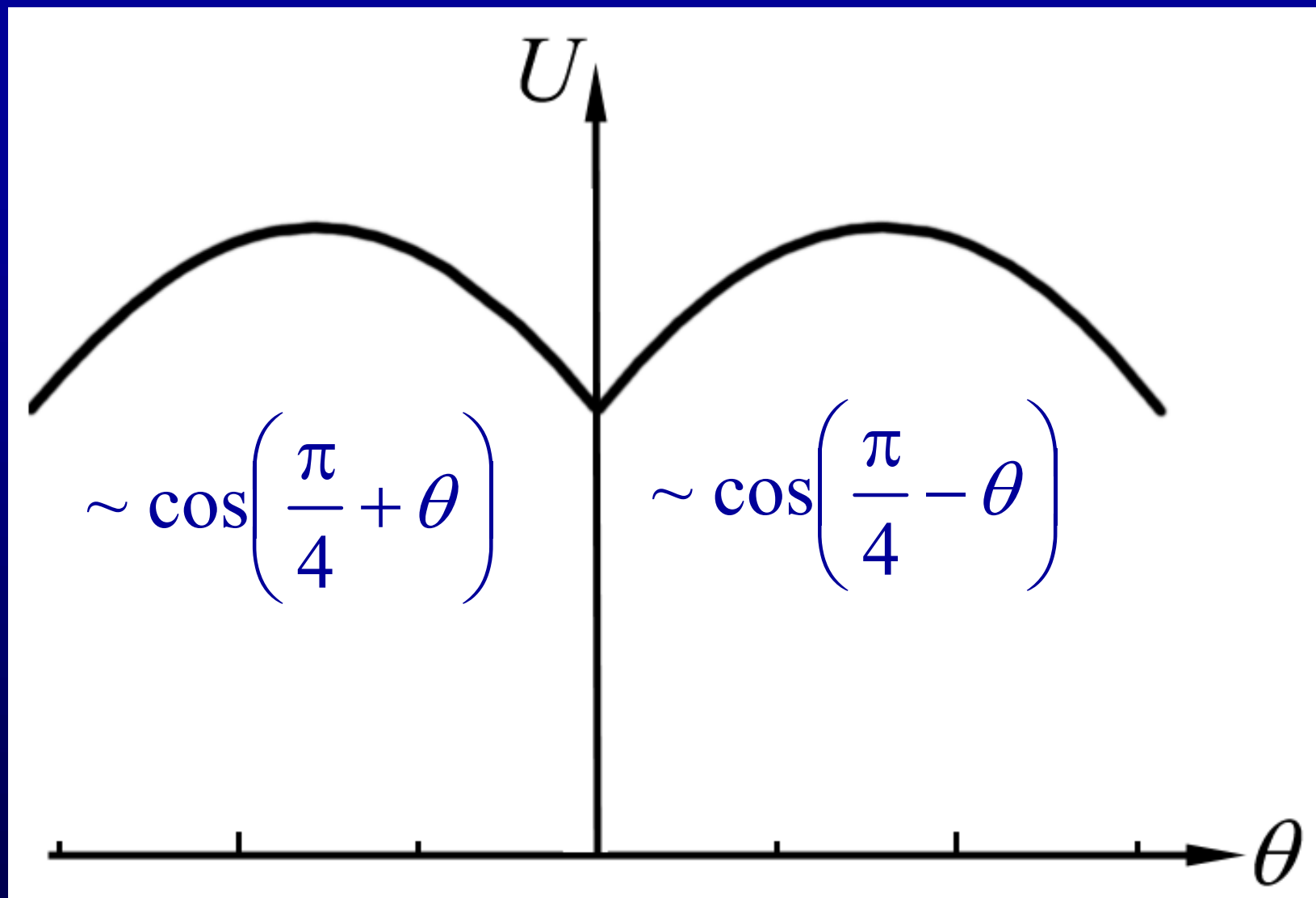
$$= \frac{1}{2} mga \left[\cos \theta + \cos \left(\frac{\pi}{2} \mp \theta \right) \right]$$

$$= mga \cos(\pi/4) \cos \left(\frac{\pi}{4} \mp \theta \right)$$

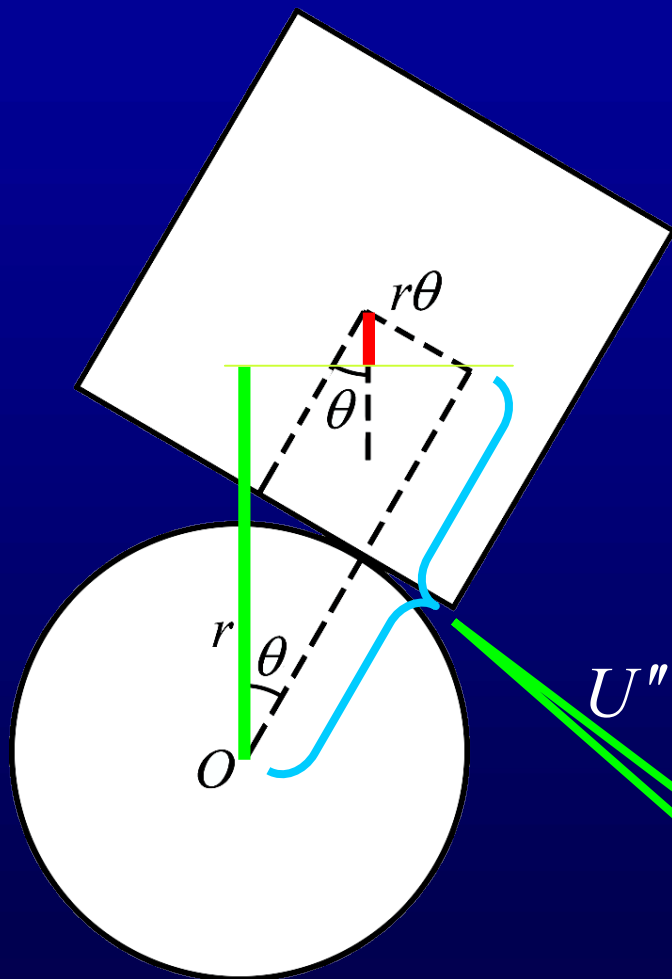
$$U'(\theta) = 0 \longrightarrow \theta = \pm \frac{\pi}{4}$$

Is $\theta = 0$ a stable equilibrium position?





Example 6.10



$$U(\theta) = mg \left[\underbrace{\left(\frac{a}{2} + r \right) \cos \theta}_{\text{green}} + \underbrace{r \theta \sin \theta}_{\text{red}} \right]$$

$$U'(\theta) = mg \left(-\frac{a}{2} \sin \theta + r \theta \cos \theta \right)$$

$\theta = 0$ is equilibrium position

$$U''(\theta) = mg \left(-\frac{a}{2} \cos \theta + r \cos \theta - r \theta \sin \theta \right)$$

$$\frac{a}{2} + r$$

$$U''(0) = mg \left(r - \frac{a}{2} \right)$$

Stable when $2r > a$

neutral when $2r = a$

unstable when $2r < a$

$$U'' = mg \left(r - \frac{a}{2} \right)$$

$$U(\theta \rightarrow 0) \Big|_{a=2r} = mga \left[\cos \theta + \frac{1}{2} \theta \sin \theta \right]$$

$$= mga \left[1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \dots + \frac{1}{2} \theta \left(\theta - \frac{\theta^3}{3!} + \dots \right) \right]$$

$$= mga \left[1 - \frac{\theta^4}{24} \right]$$

**The first nonzero
derivative is**

$$d^4 U / d\theta^4 = -mga < 0$$

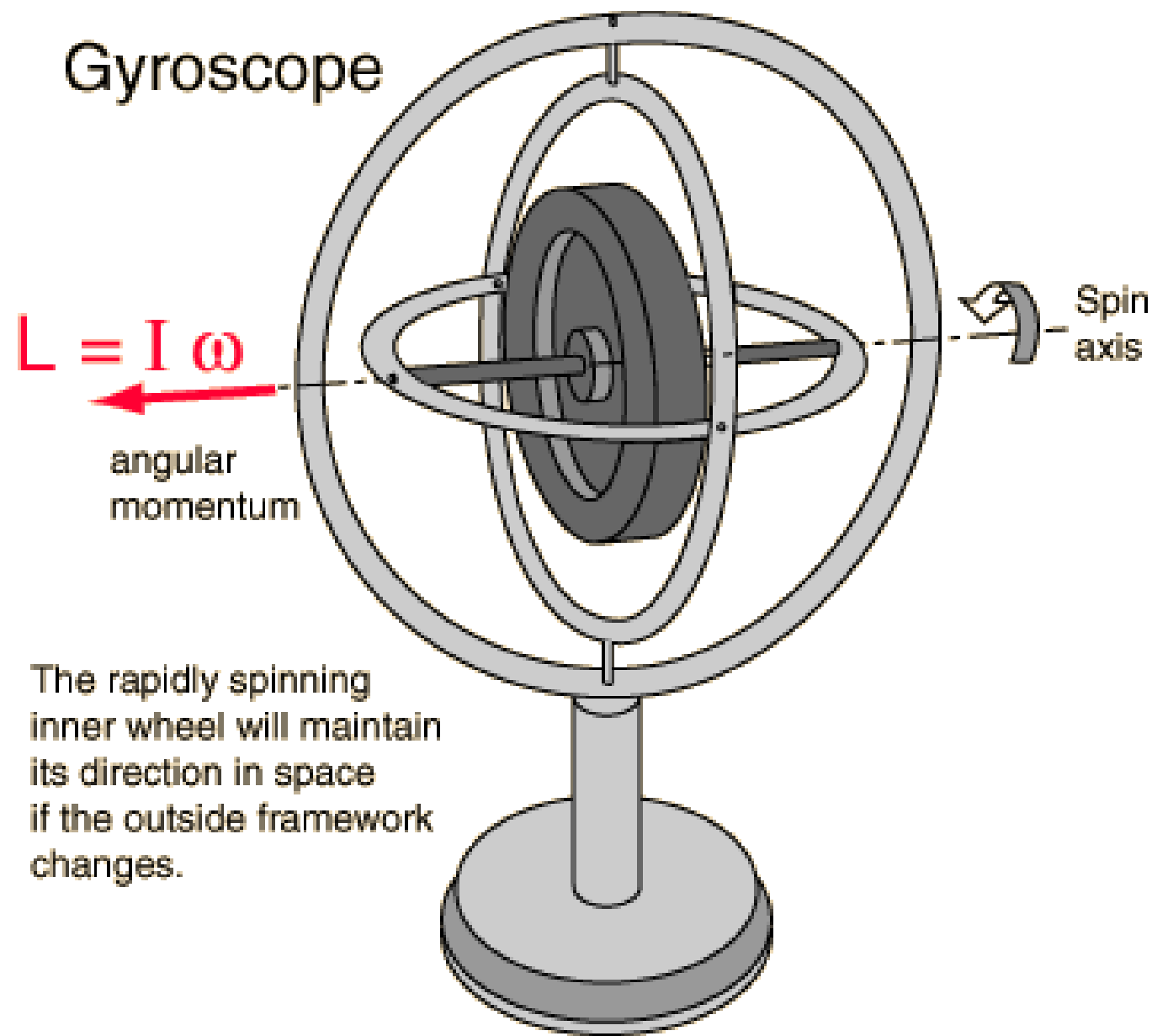
Assignment: *6.5, 6.6, 6.12

equinoxes

1. Either of two points on the celestial sphere at which the ecliptic(黄道) intersects the celestial equator(天球赤道).
2. Either of the two times during a year when the sun crosses the celestial equator and when the length of day and night are approximately equal; the vernal equinox(春分) or the autumnal equinox(秋分).

summer solstice winter solstice

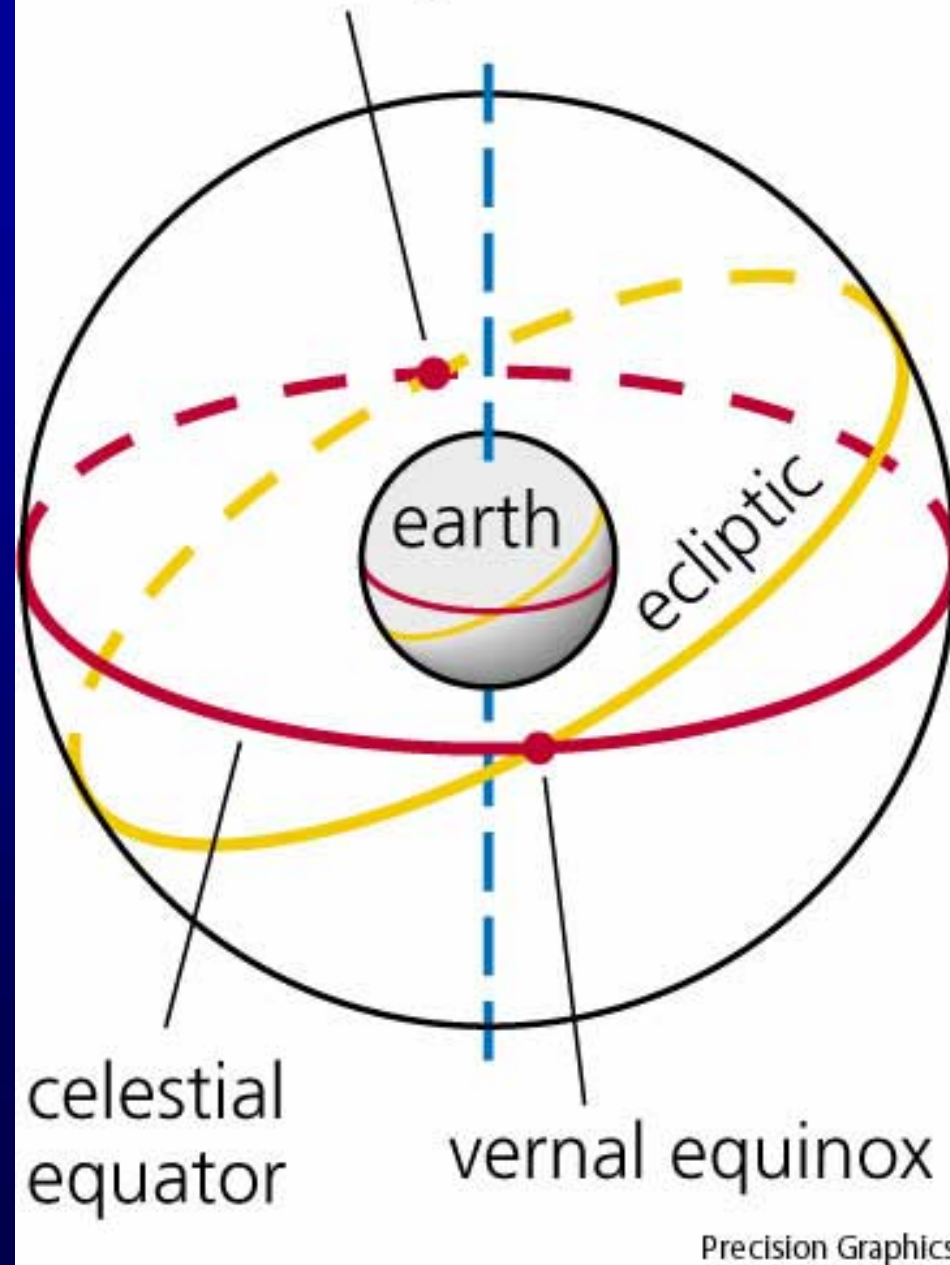
Gyroscope



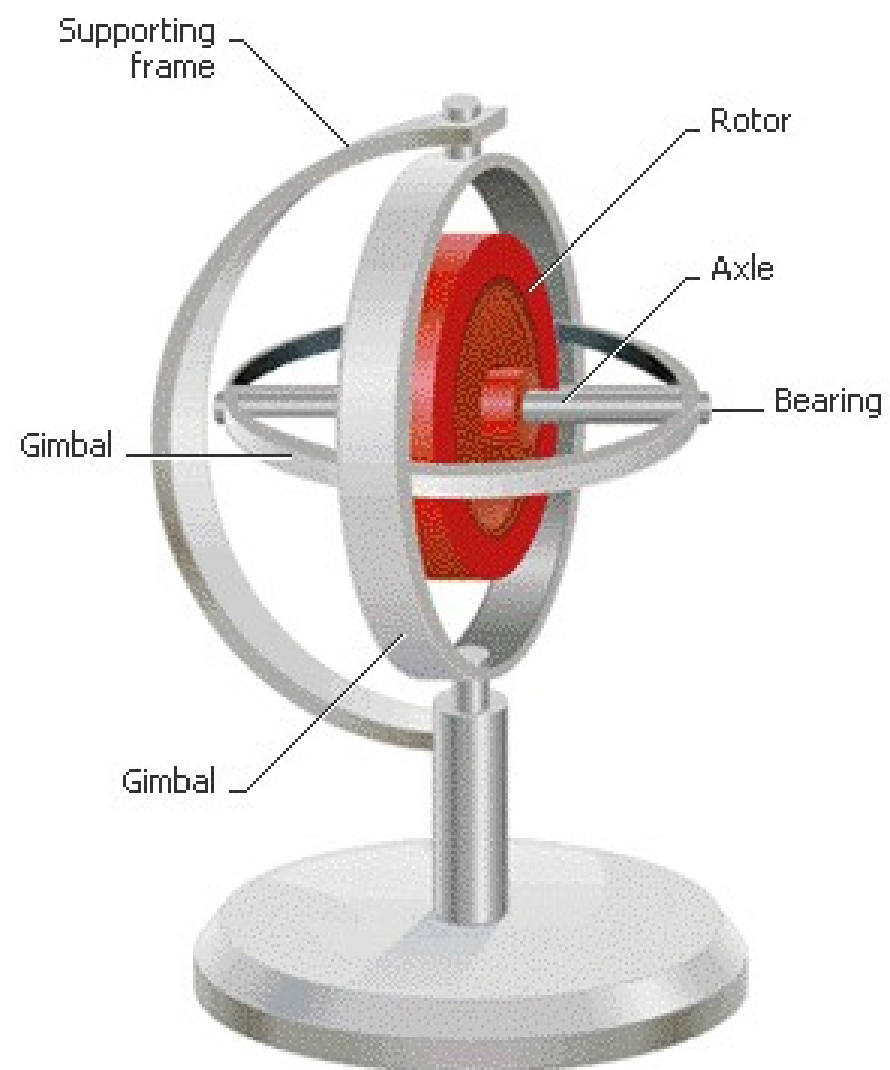
The origin of the torque on the earth:

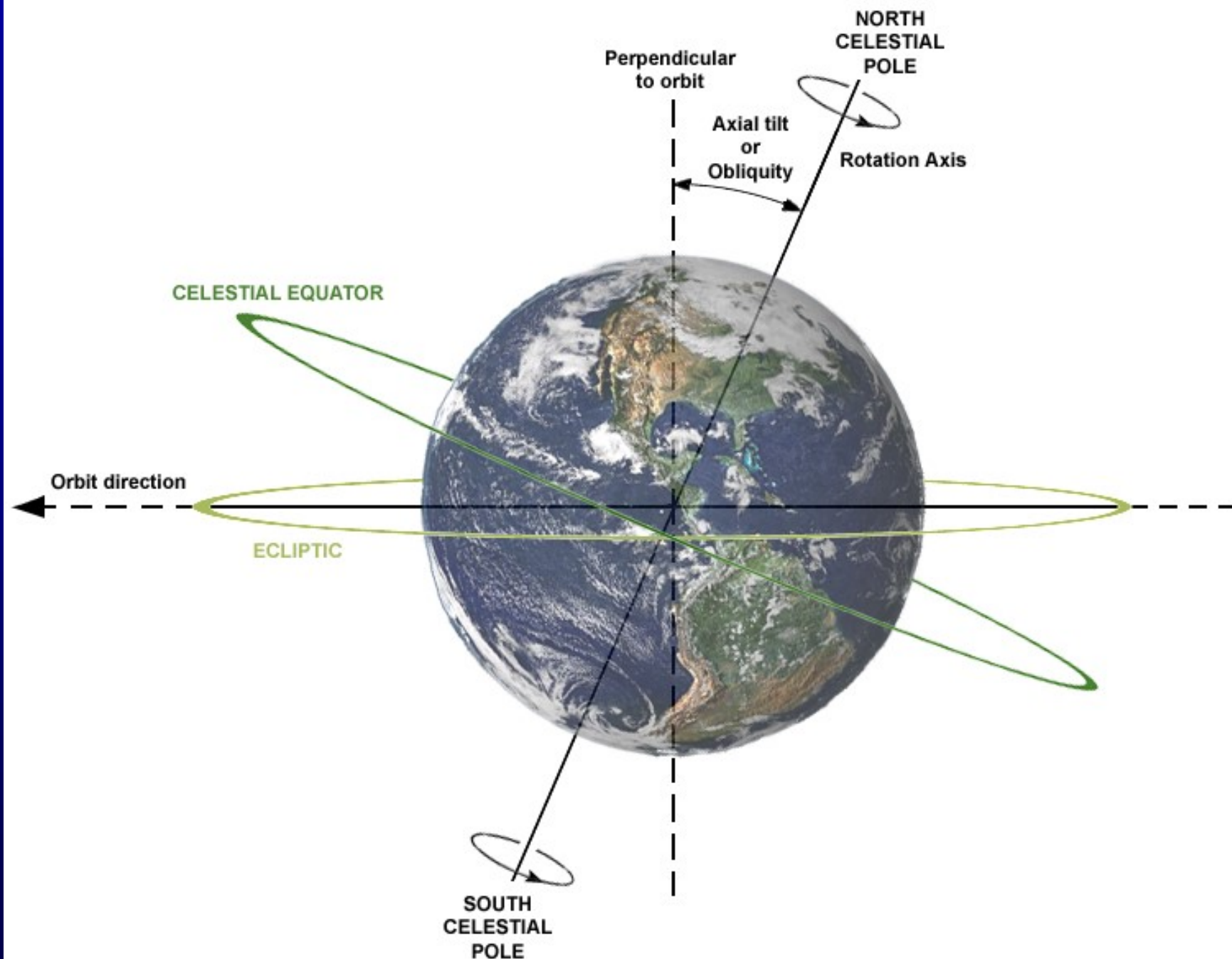
The torque which causes the Earth to precess comes from the gravitational pulls of the Sun and the Moon which try to pull the Earth's rotation axis toward the perpendicular to its orbital plane (the ecliptic plane). If the Earth were a perfect uniform sphere, there would be no precessional torque. The equatorial bulge of less than 1% plus any departures from spherical symmetry in density distribution are the only things which give the Sun and Moon any "handle" with which to exert torque on the Earth.

autumnal equinox



Precision Graphics



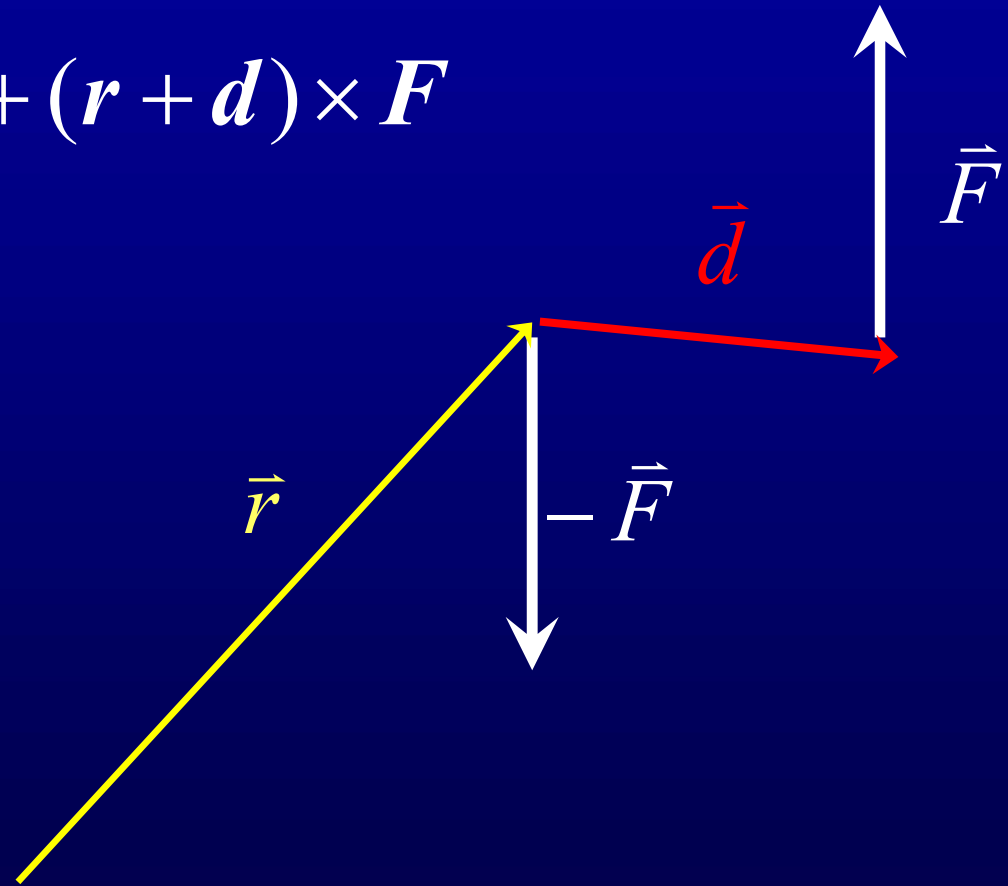


Tropic of Cancer
or Northern tropic
北回归线

Tropic of Capricorn,
or Southern tropic
南回归线

Torque of a force couple (力偶)

$$\begin{aligned} \mathbf{M} &= \mathbf{r} \times (-\mathbf{F}) + (\mathbf{r} + \mathbf{d}) \times \mathbf{F} \\ &= \mathbf{d} \times \mathbf{F} \end{aligned}$$



is independent of the reference point.